Government of
Maharashtra

"Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra on EPC Mode."

SCHEDULES

National Highway Division, Nagpur

(Ministry of Road Transport &Highways)
Government of India

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SCHEDULES

SCHEDULE-A

(See Clauses 2.1 and 8.1)

A. SITE OFTHEPROJECT

1. The Site

1.1 Site of the Four Lane Project Highway shall include the land, buildings, structures and road works as described in Annex-I of this Schedule-A.

1.2 The dates of handing over the Right of Way to the Contractor are specified in Annex-II of this Schedule-A.

1.3 An inventory of the Site including the land, buildings, structures, road works, trees and any other immovable property on, or attached to, the Site shall be prepared jointly by the Authority Representative and the Contractor, and such inventory shall form part of the memorandum referred to in Clause 8.2.1 of this Agreement.

1.4 The alignment plans of the project highway are specified in Annex-III. In the case of sections where no modification in the existing alignment of the Project Highway is contemplated, the alignment plan has not been provided. Alignment plans have only been given for sections where the existing alignment is proposed to be modified. The proposed profile of the project highways shall be followed by the contractor with minimum FRL as indicated in the alignment plan. The contractor, however, may improve/upgrade the road profile as indicated in Annex-III based on site/design requirement.

1.5 The status of the environment clearances obtained or awaited is given in Annex-IV.

Annex – I: Site

(Schedule-A)

Site for Four laning of Project Highway

1. Site

The site includes Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra. The land, carriageway and structures comprising the Site are described below.

2. Land

The Site of the Project Highway comprises the land (sum total of land already in possession and land to be possessed) as described below:

Sr. No.	Design Chainage		Existing Right of Way (m)	Proposed Right of Way (m)	Remarks
	From	**To**			
1	168+180	168+920	13.00 to 16.00	13.00 to 16.00	BSNL Office to Railway Crossing (Land Already in possession)

3. Carriageway

The present carriageway of the Project Highway is of Two Lane. The type of the existing pavement is flexible.

4. Major Bridges:

The Site includes the following Major Bridges:

Sr. No.	Existing Chainage (Km)	Type of structure			No. of Spans with span length (m)	Total Width (m)	Remarks
		Type of Foundation	Sub Structure	Super Structure			
---Nil---							

5. Road over-Bridges (ROB) / Road under-bridges (RUB):

The site includes following ROB:

S. No.	Chainage (km)	Type of Structure		No. of Spans with span length (m)	Width (m)	ROB/ RUB
		Foundation	Superstructure			
---- NIL ----						

6. Grade separators:

Sr. No.	Chainage (km)	Type of Structure		No. of Spans with span length (m)	Width (m)
		Foundation	Superstructure		
--- NIL ---					

7. Minor bridges:

Sr. No	Design Chainage (Km)	Type of Structure			No. of Spans x span length (m)	Width (m)
		Foundation	Sub structure	Super structure		
--- NIL ---						

8. Railway level crossings:

Sr. No.	Existing Road Chainage	Level Crossing no	Remark
--- NIL ---			

9. Underpasses (vehicular, non-vehicular):

Sr. No.	Chainage (km)	Type of Structure	No. of Spans with span length (m)	Width (m)
--- NIL ---				

10. Culverts

The Site has the following culverts:

(a) Slab/Box/Arch Culverts:

Sr. No	Design Chainage (Km)	Type of Culvert (Slab/Box)	No. x Span / Opening with Span Length (m)	Width (m)	
				Clear Width	Overall Width
--- NIL ---					

(b) HP Culverts:

Sr. No.	Design Chainage	Structure Type	Row X Dia. of Pipe (m)	Width (m)
1	168+200	Pipe Culvert	1x0.9	10.00
2.	168+780	Pipe Culvert	1x0.9	10.00
3.	168+910	Pipe Culvert	1x0.9	10.00

11. Bus bays and Bus Shelters:

S. No.	Chainage(km)	Length (m)	Left Hand Side	Right Hand Side
--- NIL ---				

12. Truck Lay byes:

S. No.	Chainage(km)	Length (m)	Left Hand Side	Right Hand Side
--- NIL ---				

13. Road side drains:

S. No.	Location		Type	
	From km	To km	Masonry/cc (Pucca)	Earthen (Kutcha)
--- NIL ---				

14. Major Junctions:

S. No.	Location		At grade	Separated	Category of Cross Road			
	From km	to km			NH	SH	MDR	Others
1	168+595		T	-	-	SH	-	-

15. Minor Junctions:

S. No.	Location		Type	
	From km	To km	Junction	Category of Cross Road
1	168+225		Y	Internal City Road
2	168+315		Y	Towards Muslim Library Sqaure

16. Bypasses:

S. No.	Name of bypass(town)	Chainage (km) From km to km	Length (in Km)
--- NIL ---			

17. Other structures:

[Provide details of other structures, if any.

18. Utilities:

The site includes the following utilities: -

(i) Electrical Utilities: -

Sr. No.	Name of Feeder	Substation Name	Chainage Number	Existing Utilities				
				11 KV Overhead Line (Rmt)	LT Overhead Line (Rmt)	Double Pole Structure with AB Switch (Nos.)	Distribution Transformer 200 KVA	Distribution Transformer 100 KVA
1	Bhandara Urban Feeder	33/11 KV Bhandara Urban Sub-Station	Km 168+180 to Km 168+920	1040	1862	2	2	2

Note:–Above length/numbers/scope of the existing LT/HT line shifting with pole and others electrical structures is indicative and minimum specified. The actual length/numbers/scope of the existing LT/HT line shifting with pole and others electrical structures shall be determined by the Contractor as per existing site conditions. Any increase in the length/numbers/scope specified in this Clause of Schedule B shall not constitute a Change of Scope.

(ii) Public Health Utilities (Water/ Sewage Pipe Lines): -

Sr. No.	Chainage	Water Supply Line (Length)									Sewage Line (Length)	
		CI Pipe (Rmt)	HDPE Pipe (Rmt)	DI Pipe K7 (Rmt)	Valves (Nos.)	Double Flange Bend (Nos.)	Flanged / Socket Tailpiece (No.)	All Flanged Tee	Double Socket Reducer (No.)	MJ Collour Coupling	With Pumping	With Gravity Flow
1	In Km 168+180 to Km 168+920	2085	290	-	-	-	-	-	-	-	-	-

Annexure– II:

As per Clause 8.3 (i)

(Schedule-A)

Dates for providing Right of Way of Construction Zone

The dates on which the Authority shall provide Right of Way of Construction Zone to the Contractor on different stretches of the Site are stated below:

Sr. No	From km to km	Length (m)	Existing RoW (m)	Proposed RoW (m)	Date of providing Right of Way
(1)	(2)	(3)	(4)	(5)	(6)
(i) Right of Way (Full Width)	168+180 to Km 168+920	740	13 to 16 m	13 to 16 m	On Appointed Date
(ii) Right of Way (part width) (a) Stretch (b) Stretch	NIL				
(iii) Balance Right of Way (width) (a) Stretch (b) Stretch	NIL				

Annexure- III

(Schedule-A)

Alignment Plans

(i) The existing alignment of the Project Highway shall be modified in the following sections as per the alignment plan indicated below:

(ii) The alignment of the Project Highway is enclosed in alignment plan. Finished road level indicated in the alignment plan shall be followed by the contractor as minimum FRL. In any case, the finished road level of the project highway shall not be less than those indicated in the alignment plan. The contractor shall, however, improve/upgrade the Road profile as indicated in Annex-III based on site/design requirement.

(iii) The contractor shall develop traffic signage plan as per the IRC SP 84-2019 [The Manual] and IRC 67-2015.

Annexure– IV

(Schedule-A)

Environment Clearances

The following environment clearances have been obtained : **Nil**

The following clearances are awaited: **Nil**

The MoEF clearance is not required as per new Notification of MoEF dated 22/08/2013 for this project highway.

SCHEDULE - B

(See Clause 2.1)

B. DEVELOPMENT OF THE PROJECT HIGHWAY

1. Development of the Project Highway

Development of the Project Highway shall include design and construction of the Project Highway as described in this Schedule-B and in Schedule-C.

2. Rehabilitation and augmentation shall include

Rehabilitation and Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra as described in Annex-I of this Schedule-B and Schedule-C.

3. Specifications and Standards

The Project Highway shall be designed and constructed in conformity with the Specifications and Standards specified in Annex-I of Schedule-D.

Annex – I

(Schedule-B)

Name of Work: "Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra."

1. Rehabilitation, Upgradation &Widening of the Existing Highway:

1.1 The Project Highway shall follow the existing alignment unless otherwise specified by the Authority and shown in the alignment plans specified in Annex III of Schedule-A. Geometric deficiencies, if any, in the existing horizontal and vertical profiles shall be corrected as per the prescribed standards for plain/rolling terrain to the extent land is available.

1.2 Width of Carriageway:

~~**1.2.1 Four lane Divided Carriageway with Paved shoulders of Concrete: -**~~

~~Sr. No.~~	~~Built-up stretch (Township)~~	~~Design Chainage~~		~~Length (in Km)~~	~~Standard Cross Section(in Meter)~~	~~TCS Type for Reference~~
		~~From~~	~~To~~			
~~1~~	~~2~~	~~3~~	~~4~~	~~5~~	~~6~~	~~7~~
~~NIL~~						

~~**Where: -** C.W.: Carriage Way; P.S.: Paved Shoulders; G.S.: Granular Shoulders; S.R.: Service Road; LHS – Left Hand Side; RHS – Right Hand Side; BHS – Both Hand Side.~~

~~i. The Contractor is required to fully develop the TCS given against each section mentioned above. The description in Column 6 in tentative and indicative.~~

~~ii. Cross-section at Major/Minor Bridge approaches are to be followed matching to adjoining cross-sections with suitable transition.~~

~~iii. Where cross-section changes from two lane to four lane a transition shall be provided as per the Manual (IRC: SP 84-2019 & IRC: SP 73-2018).~~

1.2.2 Two lane Carriageway with Paved shoulders : -Two lane with paved shoulders shall be constructed. The paved carriageway shall be 10 m wide in accordance with the typical cross sections drawings as per the Manual (IRC: SP 73-2018) as mentioned in the Table below.

Sr. No.	Built-up stretch (Township)	Design Chainage		Length (in Km)	Cross Section (in Meter)	TCS Type for Reference
		From	To			
1	2	3	4	5	6	7
1	Bhandara City	168+180	168+920	0.740	7.00 m (Main Carrigeway) + 2 x 1.50 m (Paved Shoulder) + Varying width of Paverblock	TCS - 1
			Total Length:	**0.740**		

Where: - C.W.: Carriage Way; P.S.: Paved Shoulders; G.S.: Granular Shoulders; S.R.: Service Road.

Note:

i. The Contractor has to fully develop the TCS-1 given above. The description in Column 6 is tentative and indicative.

ii. Where cross-section changes from two lane to four lane a transition shall be provided as per the Manual (IRC: SP 84-2019 & IRC: SP 73-2018).

iii. Cross-section at Major/Minor Bridge approaches are to be followed matching to adjoining cross-sections with suitable transition.

1.2.3 Overlay in Existing ROB section:

Sr. No.	Design Chainage		Length	Carriageway Width (in m)	TCS Type for Reference
	From	To	(In Km)		
NIL					

Note: - Mastic Asphalt with appropriate thickness shall be laid over the ROB Bridge Proper.

1.2.4 New Two Lane ROB: Approaches of New two lane RoB shall be constructed as per following cross section.

Sr. No.	Design Chainage		Length	Cross Section (in Meter)	TCS Type for Reference
	From	To	(in Km)		
NIL					

1.2.5 Except as otherwise provided in this Agreement the Linear dimension and width of the paved Carriageway and cross-sectional features shall conform to paragraph 1.2 above & Cross Sections given at Schedule.

2. Geometric Design and General Features

2.1 General

Geometric design and general features of the Project Highway shall be in accordance with Section 2 of the Manual. (IRC: SP 84-2019 & IRC: SP 73-2018).

2.2 Design speed

The design speed shall be the minimum design speed of 80/100 km per hr for plain/ rolling terrain. In urban stretch of town/village portion design speed is restricted to 40/50/65 Km per hr. where development is proposed within the extent of available land

~~**2.3 Improvement of the Existing Road Geometrics**~~

2.4 Right of Way

Details of the Right of Way are given in Annex II of Schedule-A.

Note:- No additional land will be provided than above, as such wherever required Toe wall/ Retaining wall of suitable heights shall be provided to confine free side slopes of highway embankment for main carriageway / service road etc. on either sides to accommodate the typical cross sections as given in Schedule –B within the proposed RoW without any additional cost to the Authority

2.5 Type of shoulders:

Shoulders shall be provided as per type cross sections given in Appendix B-1 of this schedule:

Sr. No.	Design Chainage		Length (in Km)	Type of Shoulders	TCS Type for Reference	Remarks
	From	To				
1	168+180	168+920	0.740	2 x 1.50 m Paved Shoulders + Avg. 2.50 m (Varying) Paverblocks/footpath)	TCS-1	Total Paverblock Area minimum 3700.00 Sq.m
		Total Length:	0.740			

Notes:

(i) Granular Shoulders shall be covered with 150 mm thick compacted layer of Crush Stone GSB.

(ii) Design and specifications of paved shoulders and granular material shall conform to the requirements specified in paragraphs 5.10 and 5.11 of the Manual. (IRC: SP 84-2019 & IRC: SP 73-2020).

(iii) The Paver block shall be Hydraulically Pressed M50 Grade of concrete and of thickness of 100 mm.

2.6 Lateral and vertical clearances at underpasses:

Lateral and vertical clearances at underpasses and provision of guardrails/crash barriers shall be as per the provision of the Manual.

Lateral clearance: The width of the opening at the underpasses shall be as follows:

Sl. No.	Location (Chainage) (Km)	Span/ opening (c/c of expansion joint) (m)	Remarks
--- NIL ---			

2.7 Lateral and vertical clearances at overpasses:

Lateral and vertical clearances at overpasses shall be as per the provision of the Manual.

Lateral clearance: The width of the opening at the overpasses shall be as follows:

Sl. No.	Location (Chainage) (Km)	Span/ opening (c/c of expansion joint) (m)	Remarks
--- NIL ---			

2.8 Service Roads

Service roads shall be constructed at the locations and for the lengths indicated below: As per IRC SP 84:2019 cl no 2.12.4

"Properly designed acceleration and declaration lanes and transition length will be considered as incidental to the project and shall not be counted towards service road length"

Sr. No.	Location of service road (Km)		Right hand side (RHS)/Left hand side (LHS)/ or Both sides	Length of service road
	From	To		
NIL				

Notes:-

(i) Above length of service/slip road is indicative and minimum specified. The actual length of the service/slip road shall be determined by the Contractor in accordance with the IRC: SP: 84:2019 requirements with approval from authority's Engineers. Any increase in the length specified in this clause of Schedule-B shall not constitute a Change of scope.

(ii) Length of service road and slip road given in above table excludes length across the Project Highway for proper connectivity of cross road on either side of Project Highway as given in the alignment plan enclosed at Annex-III, Schedule-A which shall be deemed to be included in the scope of work.

2.9 Grade separated structures:

Grade separated structures shall be provided as per provision of the Manual. The requisite particulars are given below:

Sl. No.	Location of structure	Number and length of spans (c/c of Expansion joint) (m)	Approach gradient	Proposed total width (m)	Remarks, if any
--- NIL ---					

2.10 Cattle and pedestrian underpass /overpass:

Cattle and pedestrian underpass/ overpass shall be constructed as follows: [Refer to the provision of relevant Manual and specify the requirements of cattle and pedestrian underpass/ overpass]

Sl. No.	Location	Type of crossing
--- NIL ---		

2.11 Typical cross-sections of the Project Highway

Typical cross section for the project highway is given in section 15 of schedule 'B'

3. Intersections and Grade Separators

All intersections and grade separators shall be as per Section 3 of the Manual (IRC: SP 84-2019 & IRC: SP 73-2020). Existing intersections which are deficient shall be improved to the prescribed standards.

Properly designed intersections shall be provided at the locations and of the types and features given in the tables below:

3.1 At-grade Intersections:

1. Major Junction – Nil

Sr. No.	Location of Intersection	Type of Intersection	Other features	(Approach Road to be Improved)
1.	168+595	T	Approach Road to be Reconstructed from subgrade level for 60.00 m length in Cement Concrete Pavement.	SH 355B

Rumble Strips: - The cross-road junctions as mentioned above shall be provided with Rumble strips at the approach of junctions as per IRC: 99.

2. Minor Junctions –02 Nos.

Sr. No.	Location of Intersection	Type of Intersection	Other features	(Approach Road to be Improved)
1	168+225	Y	Approach Road to be Reconstructed from subgrade level for 40.00 m length in Cement Concrete Pavement.	Internal City Road
2	168+315	Y	Approach Road to be Reconstructed from subgrade level for 40.00 m length in Cement Concrete Pavement.	Towards Muslim Library Square

Rumble Strips: - The cross-road junctions as mentioned above shall be provided with Rumble strips at the approach of junctions as per IRC: 99

The slope of intersecting cross road shall not be steeper than 1:40.

Note: -

(i) The list of above major & minor junctions are indicative only and any additional junctions required in the Project Highway shall have to be constructed by the Contractor in consultation with the Authority. However, no change of scope on account of additional junction will be accepted.

(ii) It is clarified that if any other deficient junctions with cross BT/CC roads is identified during development period in addition to those mentioned above shall be improved as per standard set forth in Schedule 'D'. Locations as above are indicative; any change in location of junction specified in the clause of Schedule 'B' shall not constitute a Change of Scope.

(iii) For cross road drainage facility, new HP culverts on cross roads shall be constructed as per Clause 7.2.4 (c) of this Schedule and shall be as per Manual.

(iv) The Contractor shall provide slab drain of sufficient capacity at all intersecting roads at which, the RCC drain is continued along the road.

4. Road Embankment and Cut Section

Widening and improvement of the existing road embankment/cuttings and construction of new road embankment. Cuttings shall conform to the Specifications and Standards given in section 4 of the Manual and the specified cross-sectional details. Deficiencies in the plan and profile of the existing road shall be corrected.

Raising of the existing road as per the provision of The Manual IRC SP 84:2019

The existing road shall be raised in the following sections:

Sl. No.	Section (from km to km)	Length	Extent of raising [Top of finished road level]
The entire stretch of project shall be raised as per enclosed Plan & Profile and as per section 4 of the manual			

Note: The chainages given in above table are indicative and stretches may increase or decrease depending upon profile designed by contractor, the raising for Reconstruction/ New Construction in approaches of Major/Minor Bridges shall be carried out by contractor as per Schedule D and in consultation with the Authority's Engineer and as per availability of land. However, this shall not be treated as change of scope.

5. Pavement Design

5.1 Pavement design shall be carried out in accordance with Section 5 of the Manual. (IRC: SP 84-2019 & IRC: SP 73-2018), IRC:58 2015, IRC:37 2018 & relevant IRC codes and International Standards.

5.2 Type of pavement

~~Flexible pavement shall be provided along the main carriageway, service roads, slip road, bus bays, truck lay bye, junction including length of cross-road (Minor Junctions) in accordance with junction improvement drawing as per IRC 37-2018.~~

Rigid pavement is proposed for the main carriageway, service roads, slip road, bus bays, truck lay bye, major junction including length of cross-road in accordance with junction improvement drawing as per IRC 58-2015.

5.3 Design requirements

Pavement design shall be done as per the section 5 of the manual and IRC: SP84:2019.

5.3.1 Design Period and strategy

Rigid pavement shall be constructed for new pavement & reconstruction of the existing road. Rigid pavement shall be designed for a design period of 30 years. Stage construction shall not be permitted.

5.3.2 Design Traffic

Notwithstanding anything to the contrary contained in this Agreement or the Manual, the Contractor shall design the pavement for design traffic of 54.83 Million Standard Axles. Considering future traffic projections for design life or as per the actual traffic whichever is higher.

The pavement composition shall not be less than the minimum pavement composition given below:

a) For Widening/Reconstruction of Rigid Pavement.

Sr. No.	Description	Minimum Crust Composition of Rigid Pavement (mm)
1	PQC	280
2	DLC	150
3	GSB	200
4	Subgrade	500
5	Embankment	As required as per site condition
	Total:	**1030**

Note: -

i) Buffer layer of 500 mm shall be provided below subgrade in the stretches of BC Soil, if occurs. However, this shall not constitute any change of scope.

ii) Crust composition mentioned above is minimum. Contractor shall execute the work as per his own design. Any increase in crust composition shall not be considered as Change of Scope

iii) The DLC shall meet the minimum cement and Compressive Strength requirement as prescribed in IRC: SP: 49. DLC will extend beyond (including that in shoulder, if any) 0.75m on either side.

iv) Below DLC layer, a properly designed drainage layer (Granular Sub Base of 200mm thickness shall be provided throughout the road width. It is designed to obtain drainage co-efficient of not less than 20 m/day.

v) A de bonding interlayer of polythene sheet having a minimum thickness as directed by the Authority Engineer is recommended for separation inter layer between DLC & PQC. Polythene sheets manufactured from recycling process shall not be allowed.

vi) Polythene sheets shall be nailed with washer as directed by Authority.

vii) Polypropylene Fibers as approved by Authority Engineer shall be used in PQC in accordance with MoRTH clause no. 602.2.5 to reduce the shrinkage cracks & post construction cracks.

~~**b) For flexible pavement (Overlay Portion) or in approach road the crust composition is:**~~

~~**Particulars**~~	~~**Name of bituminous layer**~~	~~**Description / Minimum Thickness of layer**~~
~~**Profile Correction & Patch Repairs**~~	~~Bituminous Macadam with 3.40 % bitumen of VG 40 grade is for profile and camber correction, with tack coat 3.00 Kg/10Sqm of Bitumen VG 10.~~	~~Thickness will be as per Site Requirement.~~
~~**Base / Binding Course**~~	~~Dense Bituminous Macadam with 4.50 % bitumen of VG 40 grade with tack coat 3.00 Kg/10Sqm of Bitumen VG 10.~~	~~60 mm thick.~~
~~**Wearing Course**~~	~~Bituminous Concrete with 5.40 % of bitumen VG 40 Grade and Tack coat @ 2.50Kg/10 Sqm of Bitumen VG 10.~~	~~40 mm thick.~~

~~**c) For flexible pavement (Approach Road) the crust composition is:**~~

~~**Description**~~	~~**Minimum Crust Composition for Flexible Pavement**~~
~~**BC**~~	~~40 mm~~
~~**DBM**~~	~~60 mm~~
~~**WMM**~~	~~250 mm~~
~~**GSB**~~	~~200 mm~~
~~**Subgrade**~~	~~500 mm~~

~~Total:~~	~~1050 mm~~

5.3.3 Black Cotton Soil: -

Wherever Black Cotton Soil is encountered, the same shall be removed up to 0.50m depth minimum and replaced with Murum or equivalent material as approved by Engineer-in Charge or Authority Engineer, satisfying compaction requirements as per clause no. 305.2.2.4 of section 300 of Ministry of road transport and highways (Specifications of Road and Bridge works- 5th Revision) The total length of embankment where replacement of black cotton soil may be required is incidental to the Contract Price and shall not be paid for separately.

5.3.4 Reconstruction of stretches: -

Refer to the provision of the manual and specify the stretches, if any, to be reconstructed.

All stretches of the existing road except realignment shall be reconstructed. These shall be designed as new pavement.

6. Roadside Drainage

Drainage system including surface and subsurface drains for the Project Highway shall be provided as per Section 6 of the Manual (IRC: SP 84-2019 & IRC: SP 73-2018).

a) Built Up RCC Drains and RCC Covers with footpath:- Built Up RCC Drains encased in 900 mm dia NP3 Pipe (As shown in the TCS-1) with M-20 grade concrete and RCC Covers with Manhole Covers/ Storm Water Grating, footpath, Barrier Type kerb alongside drain and paving blocks above drain slab (As shown in the TCS-1) shall be provided at following locations:

Following are the minimum lengths of drain to be constructed in the given sections. Contractor will have to rectify, redo the unfinished drain work done by previous contractor including making new water inlets, water intake chamber, finishing & dressing of drain, cleaning of debris, placing/replacing new manhole covers on open & damaged manhole covers etc all finishing work.

Sr. No.	Design Chainage		Length (M)		TCS for Reference
	From	To	L.H.S.	R.H.S.	
1	168+180	168+920	740	740	TCS - 1
			Total Length of Drain = 1480 m		

b) Earthen Drain:

Sr. No.	Design Chainage		Length (Km)		TCS for Reference
	From	To	L.H.S.	R.H.S.	
Where ever required in contract stretch.					

Note:-

i) Above length of the RCC drains are indicative and minimum specified. The actual length of the lined drains shall be determined by the Contractor keeping in view the drainage location and in accordance with the Manual requirements with approval from the Authority's Engineer. Any change in the location of drain or any other stretch where drain is required for

effective drainage as specified in above location shall not constitute a Change of Scope.

ii) RCC covered drains shall be provided with Footpath (Paving Blocks of 60 mm thickness on it & kerb stones on Road side as shown in TCS).

iii) At intersections RCC Drain shall be designed as slab drain to cater traffic matching to the standard of intersecting road.

iv) Suitable crossing raft & with thick slab of required strength shall be provided at all intersections of roads & sunken portions shall be provided to the approaches to properties etc.

v) Invert levels of drain shall be decided on the basis of ground slopes of adjoining properties and open grounds.

vi) The invert level of Roadside drains shall be minimum 500 mm below GSB bottom level & additional weep holes of 100 mm dia @1m c/c spacing shall be provided at GSB level in inner wall of built-up drain.

vii) Drain top level shall be kept such as to suit/match the surface drainage requirements as per proposed FRL in built-up sections.

viii) RCC drains on either side of the carriageway should be connected to the nearest culvert/bridge. The length and location shown above table for drain is indicative and may vary as per site condition and no change of scope shall be provided if any increase in length.

ix) The cross section of earthen drain in open country section shall have minimum bottom width of 0.5 m with side slopes of 1:1. Invert level of earthen side drains shall be as per hydraulic requirement & minimum 1 mt. below adjoining ground level.

x) Drains to be connected to the nearest culvert/bridge. The length and location shown above table for drain is indicative and may vary as per site condition and no change of scope shall be provided if any increase in length.

xi) During Execution Circular No-18.48 dated 8th July 2020 shall be strictly followed while approving drainage plan.

7. Design of Structures:

7.1 General

7.1.1 All bridges, culverts and structures shall be designed and constructed in accordance with the provision of relevant Manual and shall conform to the cross- sectional features and other details specified therein.

7.1.2 Width of the carriageway of Minor bridges structures shall be as follows:

Sr. No.	Bridge at Design Km.	Width of carriage way and Cross-Sectional features
	NIL	

Notes: -

a) All bridges shall be high-level bridges.

b) Span arrangement of Major and Minor Bridge shall be designed by contractor on the basis of detailed surveys & investigations subject to minimum specified in GAD as per Para 7.3.2 of this schedule. Founding levels shall be decided after detailed Geo-Technical investigation, Waterway, road top level, soffit etc. shall be decided on the basis of land survey conforming to various Codal provision applicable

c) The Contractor in consultation with concerned irrigation department shall plan for the construction of bridges on canal crossings considering the canal closure period. No Change of scope/ Claim shall be allowed in this aspect.

d) The Contractor shall finalize the span arrangement, design of bridges in consultation with concerned Department.

7.1.3 The following structures shall be provided with footpaths:

Sr. No.	Design Km	Type of Bridge
NIL		

7.1.4 All bridges shall be high-level bridges.

7.1.5 The following structures shall be designed to carry utility services specified in table below:

Sr. No.	Bridges @Km.	Utility Services to be carried	Type of Bridge
NIL			

Note:-The utility services may be carried outside the railing/ parapet at the bridge superstructure. The fixing and supporting arrangements with all the details shall be provided as per Manual.

7.1.6 Cross-section of the new culverts and bridges at deck level for the Project Highway shall conform to the typical cross-sections given in section 7 of the Manual. Manual (IRC: SP 84-2019 & IRC: SP 73-2020) and deviations given at Schedule D.

7.2 Culverts

7.2.1 Overall width of all culverts shall be equal to the roadway width of the approaches.

7.2.1 Reconstruction of existing culverts: -

The existing culverts at the following locations shall be re-constructed as new culverts:

Refer to paragraph 7.3 (ii) of the Manual (IRC: SP 84-2019 & IRC: SP 73-2020).

Box Culvert :- 01 Nos.

Sr. No.	Design Chainage	Existing Type of Structure	Proposed Type of structure	Proposed Span / Opening (m)
1	168+910	HPC of 1X0.9 m	Box Culvert	3.00 m

Hume Pipe Culverts :- 02 Nos.

Sr. No.	Design Chainage	Existing Type of Structure	Proposed Type of structure	Remark
1	168+200	HPC of 1X0.9 m	HPC of 1X1.2 m	Reconstruction
2	168+780	HPC of 1X0.9 m	HPC of 1X1.2 m	Reconstruction

Note: -

i. The pipes to be used for Hume Pipe Culverts should be NP-4 RCC pipe conforming to MoRTH standards and specifications.

ii. The span/opening given above are minimum and the proposed dimensions shall be based on hydraulic calculations (also depth of Box Culverts) and approval from Authority. Any change in span/opening shall not be constitute change of scope.

iii. Wherever Service Road/ Slip/ Connecting roads are proposed, the width of the culvert shall be planned beyond the Service Road/ Slip/ Connecting roads so as to ensure proper drainage of storm water outside ROW.

iv. It is clarified that as per site requirement New HP Culverts if required for drainage arrangement shall be identified & constructed if any during development shall be constructed as per standard set forth in Schedule 'D' & as per instruction of Authority.

v. The pipe diameter and numbers given above are minimum and the proposed dimensions shall be based on hydraulic calculations and approval from Authority. Any changes to above dimensions shall not constitute any change of scope.

vi. Barrel Length may change as per height of embankment and shall get approval from Authority. Any change in the length specified in this Clause of Schedule B shall not constitute any Change of Scope.

vii. Culverts should also extend to service road/slip lane as per applicable. Also, culverts shall be provided for full width as per applicable Typical Cross section.

7.2.2 Widening of existing culverts:

All existing culverts which are not to be reconstructed shall be widened to the road way width of the Project Highway as per the typical cross section given in section 7 of the Manual. Repairs and strengthening of existing structures where required shall be carried out.

Sl. No.	Culvert location	Type, span, height and width of existing culvert (m)	Repairs to be carried out [specify]
--- NIL ---			

7.2.3 Additional new culverts shall be constructed as per particulars given in the table below:

Sl. No.	Culvert Location	Proposed Span / Opening (m)
NIL		

Note: -The span/opening given above are minimum and any change in dimensions will require approval from Authority Engineer. Any change in span/opening shall not be constituting change of scope.

7.2.4 Repairs/replacements of railing/parapets, flooring and protection works of the existing culverts shall be undertaken as per Para 7.22 of manual (IRC: SP 84-2019 & IRC: SP 73-2020).

Sl. No.	Location at km	Type of repair required
--- NIL ---		

Note: - It is clarified that above repairing & strengthening measures are indicative and minimum specified. The condition survey of the existing structure shall be carried out by the contractor as per IRC 35, &strengthening / repairing shall be carried out as per the requirement of site and as per instruction of Authority Engineer, for pier, abutment, sub structure & super structure, replacement of bearings, expansion joints & wearing coat, providing railing on bridge, painting & protection works etc. If any increase in the specified above shall not constitute a Change of Scope.

7.2.5 Floor protection works shall be as specified in the relevant IRC Codes and Specifications.

7.3 Bridges:

7.3.1 Existing bridges to be re- constructed/widened:

(i) The existing bridges at the following locations shall be re-constructed as new Structures:

Sl. No.	Bridge location (km)	Salient details of existing bridge (No. of Span x Span Length x Width)	Proposed Minimum Length of structure (m)	Proposed Overall Width(m)	Type of Structure
NIL					

(ii) The following narrow bridges shall be widened:

Sl. No.	Location (km)	Existing width (m)	Extent of widening (m)	Cross-section at deck level for widening @
NIL				

7.3.2 Additional new bridges:

Sl. No.	Location (km)	Total length (m)	Remarks, if any
--- NIL ---			

7.3.3 The railings of existing bridges shall be replaced by crash barriers at the following locations:

Sl. No.	Location (km)	Total length (m)	Remarks, if any
--- NIL ---			

7.3.4 Repairs/replacements of railing/parapets of the existing bridges shall be undertaken as follows:

Sl. No.	Location (km)	Total length (m)	Remarks, if any
--- NIL ---			

7.3.5 Drainage system for bridge decks:

An effective drainage system for bridge decks shall be provided as specified in the provision of the Manual.

Note for Bridges: -

i. Span arrangement of Major and Minor Bridge shall be designed by contractor on the basis of detailed surveys & investigations subject to minimum specified in GAD. Founding levels shall be decided after detailed Geo-Technical investigation, Waterway, road top level, soffit etc. shall be decided on the basis of land survey conforming to various codal provision applicable.

ii. Where the Existing Bridge reconstructed as New Bridge at same place, dismantling of Existing Major/Minor Bridge shall be carried out with all lead and lifts. A temporary diversion works shall be constructed as per Fig. 9.25 and clause 9.9.1 of manual IRC SP 84-2019 and as per Para 112 of standards and specifications of Indian Roads Congress (MORTH) Fifth Revision-2013 with adequate cross drainage structure and traffic safety and control devices. The diversions shall be maintained in satisfactory condition till such time they are required and as directed by the Authority Engineer.

iii. Bridges should also extend to service road/slip lane as per applicable. Also, Bridges shall be provided for full width as per applicable Typical Cross section.

iv. River Training works for distance of 100 m on upstream and downstream side of bridge (total

length 200 m) shall be done by the contractor in all bridges.

v. For Widening of existing bridge: Repairs such as replacement of bearings, expansion joints & wearing coat, providing railing on bridge, painting & protection works etc. of the above mentioned Minor Bridges shall be carried out after condition survey. This shall not constitute COS.

7.4 Rail-Road Bridges:

7.4.1 Road Over Bridges:

Road over-bridge (road over rail) shall be provided at the following level crossings, as per GAD drawings attached. The details of ROB given is based on site investigation and is indicative however contractor shall study the site in details, GAD and any increase in length of ROB shall not constitute COS.

Sr. No.	Design Km.	Number and length of span (m)
NIL		

Note:

i. The proposed span arrangements of the ROBs are tentative and subject to change as per availability of railway boundaries, requirement of the railways etc.

ii. ROB shall be designed, constructed and maintained as per the requirements of Railway authorities. The construction plans shall be prepared in consultation with the concerned railway authority.

iii. The ROB shall be constructed and maintained by the Contractor under supervision of the Railways. All expenditure related to construction, maintenance and supervision of ROB (except P&E charges) shall be borne by the Contractor.

7.4.2 Grade Separated Structure: Nil

7.4.3 Repairs and strengthening of bridges and structures:

A. Bridges:

Sr. No.	Location (km)	Remarks, if any
NIL		

Note :- Following are the minimum length of bridge to be constructed in the given sections. Contractor will have to rectify, redo the unfinished bridge work done by previous contractor, finishing & dressing of Protection work, cleaning of debris including bridge to be completed in all respect as per provisions in attached drawing & standard engineering practices.

B. ROB / RUB: Nil

C. Overpasses / Underpasses and other structures: Nil

Note: - It is clarified that above repairing & strengthening measures are indicative and minimum specified. The condition survey of the existing structure shall be carried out by the contractor as per IRC 35, &strengthening / repairing shall be carried out as per the requirement of site and as per instruction of

Authority Engineer, for pier, abutment, sub structure & super structure, replacement of bearings, expansion joints & wearing coat, providing railing on bridge, painting & protection works etc.If any Increase in the specified above shall not constitute a Change of Scope.

8. RE Wall:

RE wall shall be constructed for High Embankment near Irrigation canal structure at ParseoniTown approaches at the following locations

Sr. No.	Design Chainage		Length (Rmt)	Remarks
	From	To		
NIL				

9. Traffic Control Devices and Road Safety Works:

9.1 Traffic control devices and road safety works shall be provided in accordance with Section 9 of the Manual.

9.2 Specifications of the reflective sheeting. Retro reflective sheeting should be of high intensity grade with micro-prismatic retro reflective element in accordance with ASTM standard D 4956-09 and IRC 67.

9.3 Details shall be as approved by Authority Engineer.

9.4 Reflective Traffic Safety Products/Round tree reflectors shall be provided on tree for easy visibility at night.

9.5 Solar lights blinkers shall be provided at major & minor junctions etc.

10. Roadside Furniture:

9.1 Roadside furniture shall be provided in accordance with the provisions of Section 9 of the Manual the list given below is indicative and contractor must provide additional road safety measures as per direction from Authority/AE and shall not be considered as COS.

9.2 Overhead traffic signs: The locations are mentioned in below table and the size shall be evolved based on design of sign boards given in IRC 67.

Sr. No	Locations (Design Chainage)	Type	Remark
NIL			

Note:- Numbers &Locations of overhead traffic signs &overhead Sign Gantry are indicative. Any change in numbers or locations may be based on actual site situation & approval from Authority's Engineer &shall not constitute change of scope.

9.3 Road Signages shall be provided as per relevant IRC's considering Safety of road users.

9.4 Two Stack Type Advance Direction Sign board or Map Type Advance Direction Sign board shall be provided at Major Junction at Km 168+595 in accordance to IRC 67 : 2012.

11. Compulsory Afforestation:

~~[Refer to the provision of relevant Manual for number of trees which are required to be planted by the Contractor.) Plantation of no. and species of trees per Km. shall be as per IRC SP 21-2009. Minimum avenue plantation 15mtr. c/c on both sides in Open Area except structures.~~

The number of trees which are required to be planted by the Contractor as compensatory afforestation should be as per Forest Conservation Act, **Ten times** the number of trees cut as

compensatory afforestation shall be as finalized by the Authority's Engineer.

12. Hazardous Locations

The safety barriers shall be provided at the hazardous locations as per Clause 7.17 of the Manual (IRC: SP: 84-2019). W-Beam metal crash barriers shall however be provided at all hazardous locations. All hazardous locations shall be finalized in consultation with the Authority Engineer.

a) **W-Beam metal crash barriers**-Beam metal crash barriers shall however be provided as per manual at desired locations and all Bridges& Culverts.

Sr. No.	Design Chainage	Type	Length (in Mtr)	Remarks
NIL				

Note:

As per Clause 9.7.1 W beam barrier shall be provided along all curves having radii upto 450m for complete length of curves including transitions and 20m further before and after the curve. Above length of the W-Beam metal crash barriers is indicative and minimum specified, the actual length of the W-Beam metal crash barriers shall be determined by the Contractor in accordance with the Manual requirements with approval from the Authority's Engineer. Any increase in the length and location specified in this Clause of Schedule B shall not constitute a Change of Scope.

~~**13. Rain water harvesting:**~~

14. Change of Scope:

a) The length of Structures and bridges specified hereinabove shall be treated as an approximate assessment. The actual lengths as required on the basis of detailed investigations shall be determined by the Contractor in accordance with the Specifications and Standards. Any variations in the lengths specified in this Schedule-B shall not constitute a Change of Scope, save and except any variations in the length arising out of a Change of Scope expressly undertaken in accordance with the provisions of Article 13.

b) The bidder shall submit his bid keeping in mind the possibility of descoping of work in above mentioned chainages. The reduction in contract amount will be done according to weightages given in Schedule 'H'. The contractor will not undertake any work in this length unless and until the directed by the Authority.

15. Typical Cross Section: As mentioned in above

Note 1: TCS schedule shown is indicative. It shall be read in concurrence with other features of Schedule B & C. As per site condition applicable arrangement of service road, cut / fill on LHS/RHS/BHS to be decided during execution and shall got approved from Authority or Independent/Authority Engineer. Any variation in length and TCS specified in the above table shall not constitute a Change of Scope.

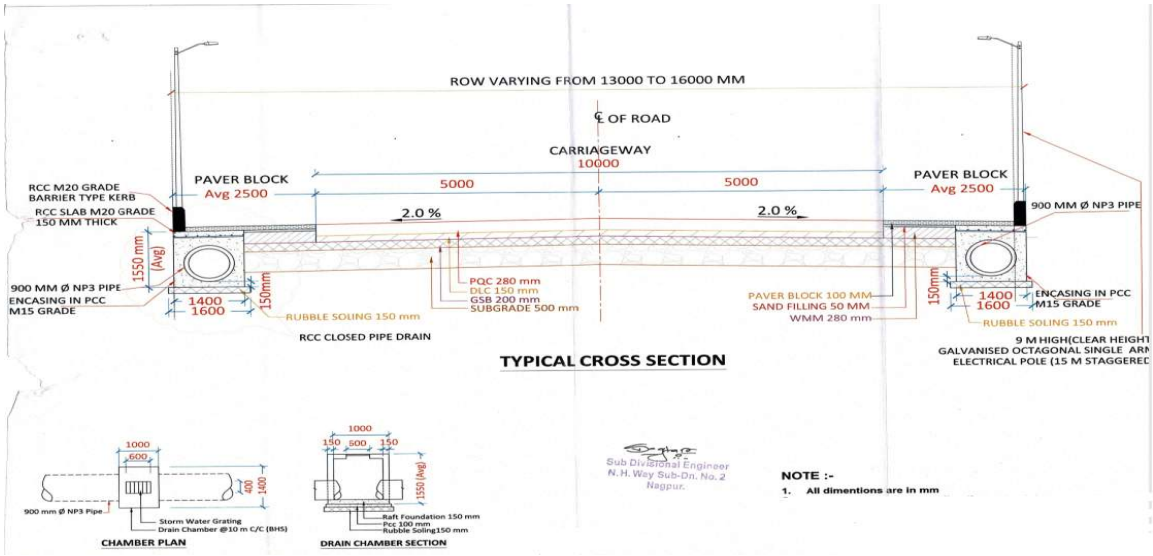
Note 2: The cross-section schedule given in above table is indicative and location of stretches within project length may change depending upon profile designed by contractor however, this shall not be treated as change of scope.

Note 3:

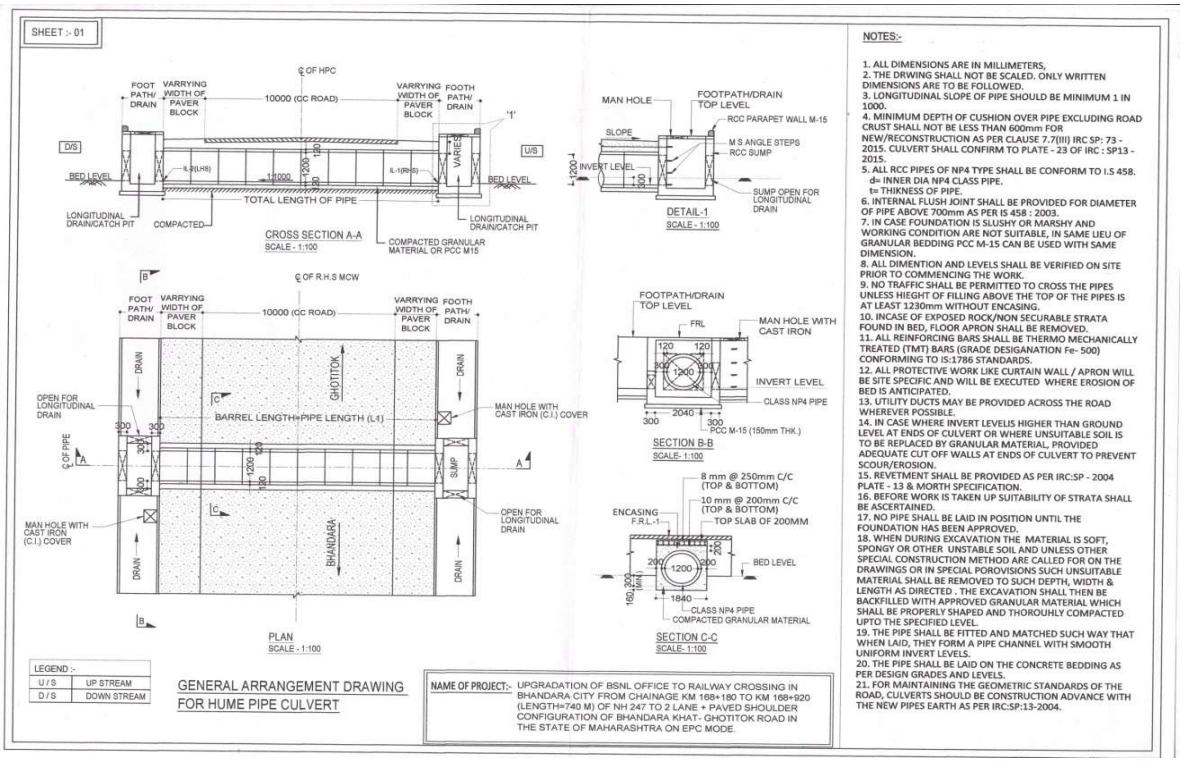
i. Cross-section at Major/Minor Bridge approaches are to be followed matching to adjoining cross-sections with suitable transition.
ii. Where cross-section changes from two lane to four lane a transition shall be provided as per the Manual (IRC: SP 84-2019 & IRC: SP 73-2018).
iii. Where Bus Bays & Truck Lay Byes constructed as per Schedule 'C' a transition shall be provided as per the Manual (IRC: SP 84-2019 & IRC: SP 73-2018).

Typical Cross Sections

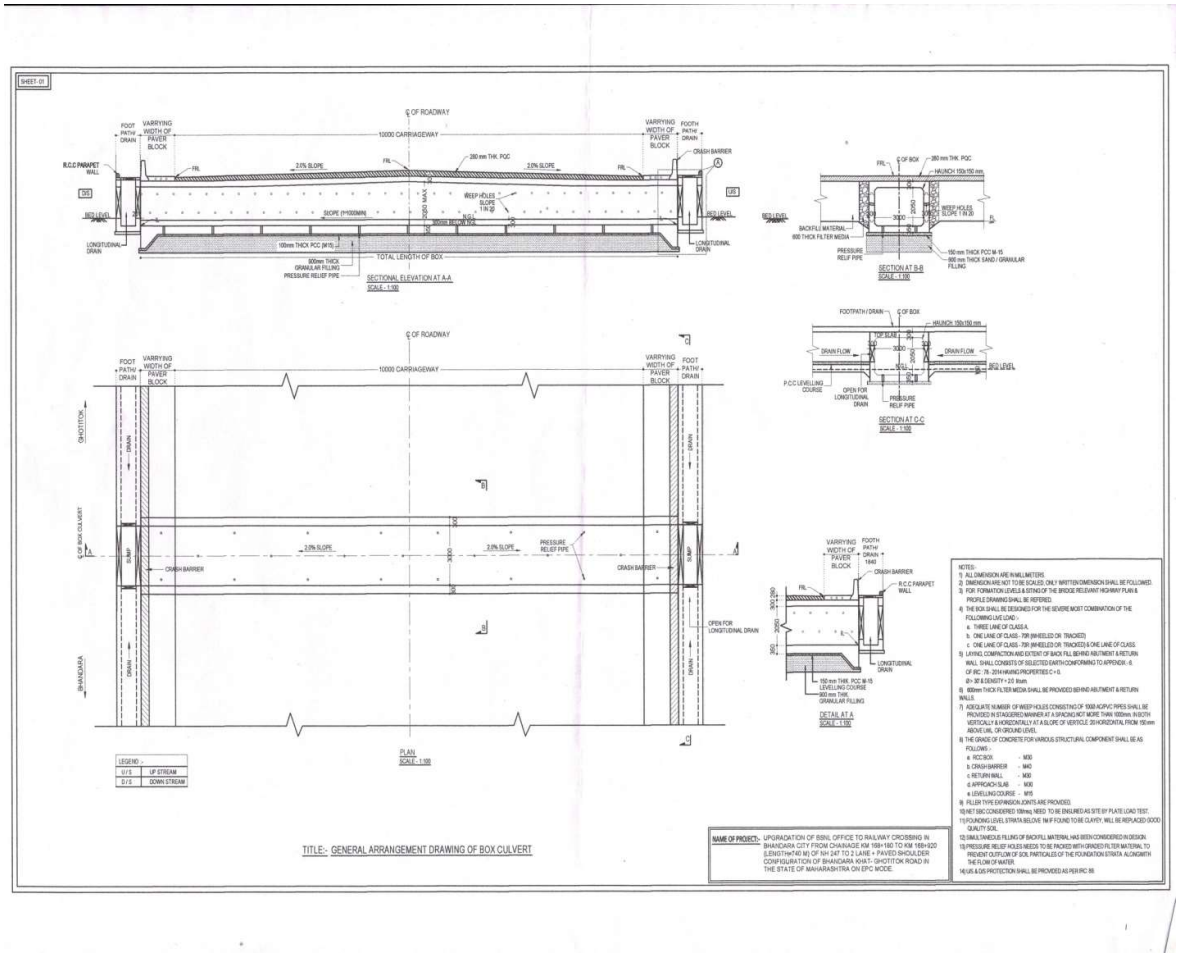
TCS Type - 1



HPC DRAWING



BOX CULVERT


SHEET-01
℄ OF ROADWAY
10000 CARRIAGEWAY
VARRYING WIDTH OF PAVER BLOCK
FOOT PATH / DRAIN
CRASH BARRIER
R.C.C PARAPET WALL
2.0% SLOPE
280 mm THK. PCC
WEEP HOLES SLOPE 1 IN 20
SLOPE (1:1000MIN)
LONGITUDINAL DRAIN
100mm THICK PCC (M15)
900mm THICK GRANULAR FILLING
PRESSURE RELIEF PIPE
TOTAL LENGTH OF BOX
SECTIONAL ELEVATION AT A-A
SCALE - 1:100
℄ OF BOX
HAUNCH 150x150 mm
BACKFILL MATERIAL
600 THICK FILTER MEDIA
150 mm THICK PCC M-15
600 mm THICK SAND / GRANULAR FILLING
SECTION AT B-B
SCALE - 1:100
FOOTPATH / DRAIN
DRAIN FLOW
P.C.C LEVELLING COURSE
OPEN FOR LONGITUDINAL DRAIN
SECTION AT C-C
SCALE - 1:100
GHOTITOK
BHANDARA
℄ OF BOX CULVERT
SUMP
OPEN FOR LONGITUDINAL DRAIN
PLAN
SCALE - 1:100
LEGEND
U/S UP STREAM
D/S DOWN STREAM
DETAIL AT A
SCALE - 1:100
NOTES:
1) ALL DIMENSION ARE IN MILLIMETERS.
2) DIMENSION ARE NOT TO BE SCALED, ONLY WRITTEN DIMENSION SHALL BE FOLLOWED.
3) FOR FORMATION LEVELS & SITING OF THE BRIDGE RELEVANT HIGHWAY PLAN & PROFILE DRAWING SHALL BE REFERRED.
4) THE BOX SHALL BE DESIGNED FOR THE SEVERE MOST COMBINATION OF THE FOLLOWING LIVE LOAD :-
a. THREE LANE OF CLASS A.
b. ONE LANE OF CLASS - 70R (WHEELED OR TRACKED)
c. ONE LANE OF CLASS - 70R (WHEELED OR TRACKED) & ONE LANE OF CLASS
5) LAYING, COMPACTION AND EXTENT OF BACK FILL BEHIND ABUTMENT & RETURN WALL SHALL CONSISTS OF SELECTED EARTH CONFORMING TO APPENDIX - 6 OF IRC 78 - 2014 HAVING PROPERTIES C = 0, Ø= 30° & DENSITY = 20 kn/m
6) 600mm THICK FILTER MEDIA SHALL BE PROVIDED BEHIND ABUTMENT & RETURN WALLS.
7) ADEQUATE NUMBER OF WEEP HOLES CONSISTING OF 100Ø ACP/VC PIPES SHALL BE PROVIDED IN STAGGERED MANNER AT A SPACING NOT MORE THAN 1000mm IN BOTH VERTICALLY & HORIZONTALLY AT A SLOPE OF VERTICLE 20 HORIZONTAL FROM 150mm ABOVE UWL OR GROUND LEVEL.
8) THE GRADE OF CONCRETE FOR VARIOUS STRUCTURAL COMPONENT SHALL BE AS FOLLOWS :-
a. RCC BOX - M30
b. CRASH BARRIER - M40
c. RETURN WALL - M30
d. APPROACH SLAB - M30
e. LEVELLING COURSE - M15
9) FILLER TYPE EXPANSION JOINTS ARE PROVIDED.
10) NET SBC CONSIDERED 100kn/m2, NEED TO BE ENSURED AS SITE BY PLATE LOAD TEST.
11) FOUNDING LEVEL STRATA BELOVE 1M IF FOUND TO BE CLAYEY, WILL BE REPLACED GOOD QUALITY SOIL.
12) SIMULTANEOUS FILLING OF BACKFILL MATERIAL HAS BEEN CONSIDERED IN DESIGN.
13) PRESSURE RELIEF HOLES NEEDS TO BE PACKED WITH GRADED FILTER MATERIAL TO PREVENT OUTFLOW OF SOIL PARTICALES OF THE FOUNDATION STRATA ALONGWITH THE FLOW OF WATER.
14) U/S & D/S PROTECTION SHALL BE PROVIDED AS PER IRC 89.
TITLE:- GENERAL ARRANGEMENT DRAWING OF BOX CULVERT
NAME OF PROJECT:- UPGRADATION OF BSNL OFFICE TO RAILWAY CROSSING IN BHANDARA CITY FROM CHAINAGE KM 168+180 TO KM 168+920 (LENGTH=740 M) OF NH 247 TO 2 LANE + PAVED SHOULDER CONFIGURATION OF BHANDARA KHAT- GHOTITOK ROAD IN THE STATE OF MAHARASHTRA ON EPC MODE.

(Schedule B-1)

DETAILS OF UTILITIES AND FELLING OF TREES

1. The shifting of utilities and felling of trees required for construction of project highway shall be carried out by the Contractor. The cost of the same including all departmental charges excluding supervision charges shall be borne by the Contractor. The shifting of the utilities shall be done as per norms of the concerned user agency and subject to their approval. The details of existing utilities at site are given in table below:

Sr. No.	Name of Feeder	Substation Name	Chainage Number	Existing Line					Proposed Line					
				11 KV Overhead Line (Rmt)	LT Overhead Line (Rmt)	Double Pole Structure with AB Switch (Nos.)	Distribution Transformer 200 KVA	Distribution Transformer 100 KVA	11 KV Underground Line (Double Run) (Rmt)	LT Underground Line (Double Run) (Rmt)	Double Pole Structure with AB Switch (Nos.)	SPCP for Cable support (Nos.)	Distribution T/F 200 KVA (Shifting) (Nos.)	Distribution T/F 100 KVA (Shifting) (Nos.)
1	Bhandara Urban Feeder	33/11 KV Bhandara Urban Sub-Station	Km 168+180 to Km 168+920	1040	1862	2	2	2	1040 X 2 = 2800	1862 X 2 = 3724	2	7	2 (Shifting Only)	2 (Shifting Only)

(ii) Public Health Utilities (Water/ Sewage Pipe Lines): -

Sr. No.	Chainage	Water Supply Line (Length)									Sewage Line (Length)	
		CI Pipe (Rmt)	HDPE Pipe (Rmt)	DI Pipe K7 (Rmt)	Valves (Nos.)	Double Flange Bend (Nos.)	Flanged / Socket Tailpiece (No.)	All Flanged Tee	Double Socket Reducer (No.)	MJ Collour Coupling	With Pumping	With Gravity Flow
1	In Km 168+180 to Km 168+920	2085	290	-	-	As directed by Utility Owning Department {Municipal Council Bhandara & Maharashtra Jeevan Pradhikaran (MJP) }					-	-

Note:

(a) All existing utility lines (electrical, water supply, sewage lines etc) located along the length of the project highway are required to be shifted to the edge of the proposed ROW at a suitable location as per directions of the Authority. The entire formation width is to be maintained clear of any utility lines.

(b) All existing utility lines (electrical, water supply, sewage lines etc) located across the length of the project highway are required to be relocated / shifted at suitable crossings through utility ducts whose locations shall be decided in consultation with the Authority and user agencies concerned.

(c) The quantities given in the table above are as per the assessment of the Authority. There may be variations in the above quantities due to various factors including buried lines which might be encountered during excavation works. Shifting of all existing surface, buried and overhead utility lines are to be shifted by the Contractor within the scope of the Contractor. No change of scope shall be accepted on account of any change in quantities / specifications of the utility lines to be shifted.

Utility Shifting Sheet-II (Annexure-I to Schedule-B 1)

Note: - Shifting of obstructing existing utilities indicated in schedule A to an appropriate location in accordance with the standards and specifications of concert Utility Owing Department is part of the scope of work of the contractor. The bidders may visit the site and access the quantum of shifting of utilities for the projects before submission of their bid. Copy of utility relocation plan is enclosed. The specifications of concerned utility owing department shall be applicable and followed.

(d) All existing utility lines (electrical, water supply, sewage lines etc) located along the length of the project highway are required to be shifted to the edge of the proposed ROW at a suitable location as per directions of the Authority. The entire formation width is to be maintained clear of any utility lines.

(e) All existing utility lines (electrical, water supply, sewage lines etc) located across the length of the project highway are required to be relocated / shifted at suitable crossings through utility ducts whose locations shall be decided in consultation with the Authority and user agencies concerned.

(f) The quantities given in the table above are as per the assessment of the Authority. There may be variations in the above quantities due to various factors including buried lines which might be encountered during excavation works. Shifting of all existing surface, buried and overhead utility lines are to be shifted by the Contractor within the scope of the Contractor. No change of scope shall be accepted on account of any change in quantities / specifications of the utility lines to be shifted

(g) The type/spacing/size/specifications of poles/towers/line/cables to be used in shifting work are as per the guidelines of utility owning department and it is solely between Contractor and the Utility owning department. No change of scope shall be admissible and no cost shall be paid for using different type/spacing/size/specifications in shifted work in comparison to

those in the existing or for making any overhead crossings to underground as per requirement of utility owning department/construction of project highway. The Contractor shall carry out joint inspection with utility owning department and get the estimates sanctioned from utility owning department. The assistance of the Authority is limited to giving forwarding letter on the proposal of Contractor to utility owning department whenever asked by the Contractor. The decision/approval of utility owning department shall be binding on the Contractor.

(h) The supervision charges at the rates/ charges applicable of the utility owning department shall be paid directly by the Authority to the Utility Owning department as and when Contractor/Concessionaire* furnishes demand of Utility Owning Department along with a copy of estimated cost given by the later.

(i) The dismantled material/scrap of existing Utility to be shifted/dismantled shall belong to the Contractor who would be free to dispose-off the dismantled material as deemed fit by them unless the Contractor is required to deposit the dismantled material to utility owning department as per Norm & Practice. In that case the amount of credit for dismantled material may be availed by the contractor as per estimate agreed between them.

(j) The utilities shall be handed over after shifting work is completed to Utility Owning Department after due testing as desired by them &upto their entire satisfaction. The maintenance liability shall rest with the Utility Owning Department after handing over process is complete as far as utility shifting works are concerned.

(k) The above scope of electrical shifting involves all the necessary fixtures & fastenings as per site condition, specification & standard practice of utility owning entity. The present available guidelines on shifting of electrical utility issued by Chief Engineer, Maharashtra State Electrical Distribution Co. Ltd. Vide Ref. No. C.E. (Dist.)/D-III/07500 Dated 15th March 2019& New Guidelines issued afterwards shall to be scrupulously followed by the Contractor.

(l) The above quantities may vary depending upon actual alignment & site conditions, hence above quantities are subjected to negative CoS.

(m) The EPC Contractor & Authority Engineer shall try to optimize the quantum of shifting. Actual length & location of shifting will be decided by Joint Inspection with Authority Engineer & Utility Owning Entity.

(n) The above scope of water shifting involves all necessary sluice valves. Air Valves, Connections, Lowering, fixing, jointing fixtures & fastenings as per guidelines & specifications of utility owning entity. The contractor shall Provide utility duct of RCC NP4 pipes of 300/600/900 mm Pipe as required for water supply underground crossing.

(o) The Utilities shall be shifted to extreme edge of RoW as per relevant TCS & as per directed by Authority Engineer.

SCHEDULE - C

(See Clause 2.1)

C. PROJECT FACILITIES

I. Project Facilities

The Contractor shall construct the Project Facilities in accordance with the provisions of this Agreement. Such Project Facilities shall include:

a) Roadside furniture
b) Pedestrian facilities
c) Landscaping & Tree Plantation
d) Bus Shelters
e) Street lighting and High mast lighting
f) Others

II. Description of Project Facilities

Each of the project facilities are briefly described below.

a) Roadside Furniture shall be provided as follows:

i. Traffic signs and pavement markings:

Traffic signs and pavement markings shall include road side signs, overhead signs, curve mounted signs and road marking along the project highway. Aluminum Composite Material (ACM) sheets as per ASTM D Type-XI shall be used for sign boards. The locations for these provisions shall be finalized in consultation with Authority's Engineer and as per latest IRC Standard.

ii. Concrete Crash Barrier, W beam crash barrier, Pedestrian guard rail (MS railings):

a) Concrete crash barrier shall be provided shall be finalized in consultation with Authority's Engineer.

b) The W-Beam Metal crash barriers shall be provided as per Schedule D and for safety of traffic & users. The locations shall be finalized in consultation with Authority's Engineer.

iii. Traffic Safety Devices in consultation with Authority's Engineer & Latest IRC Standards.

iv. Boundary Stones shall be placed throughout the project road as per IRC SP-73-2018, Clause 12.2 and latest guidelines of IRC 25.

v. Hectometer/kilometer Stones as per IRC SP-73-2018, Clause 12.3 and latest guidelines of IRC 8.

vi. Solar Traffic blinker signal (L.E.D) shall be provided at intersections (Minimum 6 Nos).

vii. Rumble strips shall of Thermoplastic paint shall be provided at all cross-road junction as per latest IRC:99

viii. Providing and fixing Steel Railing complete as per drawing and technical specifications section 800, wherever required as per site requirements in consultation with Authority Engineer. Length of Steel railing provided shall not be claimed and considered under change of scope.

b) Pedestrian facilities -

The additional pedestrians' facilities in the form of guard rails, footpath, lighting etc. shall be provided in built-up area and high embankment of structures.

~~For proper approaches to adjoining agricultural fields along excavated drains, 900mm dia. NP-4 pipes for about 7.5m in length with adequate cushion over pipe at an interval of 1000 mtr on both sides & wherever required as per site requirements in consultation with Authority Engineer shall be provided.~~

~~**c) Landscaping and Tree Plantation:**~~

~~**Roadside Tree Plantation:**~~

Sr. No.	Design Chainage (km)	Side	Total No. of Trees
NIL			

~~**Small flowery plants / Grass turf on median**~~

S. No.	Design Chainage (km)	Provision
NIL		

d) Bus Shelters:

Bus Shelter shall be provided at locations given below. The bus bay and shelter shall be designed based on IRC SP-84-2019, Clause 12.6 and figure 12.3

Sr. No.	Village Name	Remarks
NIL		

Note: The locations of Bus shelters are tentative & shall be got approved / provided in consultation with the Authority / Authority's Engineer.

e) Street Lighting & High Mast Lighting:

i. Street/Highway Lighting:

Street Light: Street lighting on octagonal lamp post with LED /energy efficient lighting system of standard make with minimum 40 Lux capacities shall be provided @ 15 m interval (staggered) for the chainages mentioned in following table. Street lights shall be provided with single light on single pole on both side of road @ 15 m interval (staggered). The height of street light pole shall be about 9 m above FRL.

Sr. No.	Chainage		Length (m)	Remarks
	From	To		
1	168+180	168+920	**740**	Single Arm Street Light on either side of road @ 15 m interval (staggered).

ii. High Mast Lighting (25 m clear height) : - 1 No. at Major Junction @ Km 168+595 (Kham Talao Sqaure)

iii. The lighting work shall be got done from the qualified specialized agency.

iv. The scope includes providing entire lighting systems, trenching, underground / building in cabling, transformers etc and obtaining electric supply / approval from concern Govt. department such as MSEDCL etc.

v. The scope includes arrangements of procuring power supply to ensure uninterrupted lighting during night and when visibility is low.

vi. The scope includes all cost of procurement and installation.

Note:-Above length considered for providing street lights is indicative and minimum specified. The actual length shall be determined by the Contractor in accordance with the Manual requirements with approval from the Authority's Engineer. Any increase in length specified in this Clause of Schedule C shall not constitute a Change of Scope.

f) Others

(i) Slope protection:

a. The slope protection by laying, pitching (minimum 300 mm thick) on slopes, laid over the prepared bank including filter media, boulder, apron (minimum 3600 mm thick) laid dry in front of toe of embankment will be provided at locations where embankment height is more than 1.00 mt. & in approaches of viaducts/flyovers/bridges as per Manual and as directed by Authority. The Indicative Locations are as follows:

b. The slope protection by turfing, vegetation or any other method using green technology will be provided at locations, where embankment height is more than 3 m and in approaches of Viaducts / Flyover / bridges as per Manual and as directed by Authority.

(ii) Utility pipe ducts

- The ducts shall be laid as per IRC-SP-84-2019, Clause 2.16 (with chamber size 900mm X 900 mm) at a suitable depth as approved by Authority's Engineer. For the purpose of utility duct provision the Built Up Area shall be the Stretch of road where RCC Box Drains are constructed.
- ~~For the area other than Built Up Section, the utility ducts of 600 mm dia. at the interval of 500 meter shall be provided.~~
- For crossing overhead electrical utility lines, at every location 2 rows of 300 mm diameter NP2 pipe at every electrical overhead crossing shall be provided (Minimum 15

Locations/Crossings). This utility ducts shall be in addition to utility ducts mentioned in above paras.

- The locations of utility duct crossing shall be finalized as per requirement of Electrical & Water supply owning entity.
- Utility Ducts shall be constructed with proper inlet and chamber for crossing service lines such as irrigation pipe lines and cables. In urban areas, the ducts shall be constructed along the project road for linear underground utility lines. The ducts shall be laid at a suitable depth as approved by Authority Engineer.

(iii) Utilities

Utilities to be identified at site and certified by the Authority Engineer then shifting may be taken by contractor.

~~**(iv) Rain Water Harvesting**~~

Note:

a) The locations/ chainage mentioned above for project facilities is indicative, may get changed and shall be finalized in consultation with the Authority's Engineer and as per availability of land, however this shall not treated as Change of Scope.

b) In case of any discrepancy in numbers or locations of any of the project facilities mentioned in this Schedule C, shall be constructed and provided in consultation with the Authority Engineer as per site/design requirement and shall not Constitute as Change of Scope.

SCHEDULE - D

(See Clause 2.1)

D. SPECIFICATIONS AND STANDARDS

1. Construction

The Contractor shall comply with the Specifications and Standards set forth in Annex- I of this Schedule-D for construction of the Project Highway.

2. Design Standards

i. The Project Highway including Project Facilities shall conform to design requirements set out in the following documents:

ii. Manual of Specifications and Standards for Two Laning of Highways (IRC:SP:73-2018),for Four Laning of Highways (IRC: SP: 84-2019) and for maintenance, Repair and Rehabitation of Cement Concrete Pavements (IRC: SP: 83-2018) referred herein as the Manual.

iii. The use of Manufactured aggregates in National Highways Works shall be governed by Ministry's Circular No. RW/NH-34066/09/2017-S&R (B) Dt. 21.07.2020.

iv. The provision of mandatory use of factory manufactured precast concrete elements in project within 100 km radius of the precast factory shall be governed by Ministry's Circular No. RW/NH-34049/01/2020-S&R(B) dated 08.04.2022. The minimum mandatory usage should be 25% of the total concrete volume other than the foundations and substructures of bridges/via duct/ ROB. Any relaxation may be granted by Authority only if the contractor/concessionaire is able to demonstrate that the manufacturer(s) is not able to provide such products matching to the works programme of the contractor or the prices demanded by the manufacturer is more than the prices derived for cast-in-situ condition at the rates of SoR prevailing on 28th days before the bid due date.

v. The use of fly ash in road/ flyover embankment construction on national highways work shall be governed by Ministry's Circular No. RW/NH-33044/01/2019-S&R(P&B)RSCE-Pt.1 dated 18.04.2022 and guidelines/ notification issued by MoEF& CC and MOP and applicable guidelines of Indian IRC & BIS.

vi. The use of UHPFRC shall be governed by Ministry's Circular No. RW/NH-33049/01/2020-S&R (B) Pt.

Annex – I: Specifications and Standards of The Project

(Schedule-D)

1. Specifications and Standards

I. All Materials, works and construction operations shall conform to the Manual of Specifications and Standards for Two-Laning of Highways IRC: SP:84-2019, referred to as the Manual, and MORTH Specifications for Road and Bridge Works. Where the specification for a work is not given, good industry practice shall be adopted to the satisfaction of the Authority Engineer.

II. As regards, the work of utility shifting, the relevant specification, relevant rules, regulations and acts of Utility Owning Department / Agencies shall be applicable. **Guidelines on shifting of electrical utility issued by Chief Engineer, Maharashtra State Electrical Distribution Co. Ltd. Vide Ref. No. C.E. (Dist.)/D-III/07500 Dated 15th March 2019 & subsequent Guidelines will be applicable for electrical shifting works.** (attached with the agreement for reference)

III. All tree plantation Materials, works and construction operations shall conform to the Guidelines for landscaping and tree plantation IRC: SP 021: -2009, referred to as the Manual, and MORTH Specifications for Road and Bridge Works. Where the specification for a work is not given, good industry practice shall be adopted to the satisfaction of the Authority Engineer.

IV. The condition of Road should be recorded by Network Survey Vehicle (NSV) before start of work, during construction, after completion & at the end of defect liability period by the Contractor under supervision of Authority Engineer and PWD (NH) Division. These surveys shall be incidental to the work and shall not be paid for extra. If Contractor does not undertake this activity at his own, then it will be got done from other agency & the payment for which will be recovered from his due payment.

2. Deviations from the Specifications and Standards

The terms "Concessionaire", "Independent Engineer" and "Concession Agreement" used in the Manual shall be deemed to be substituted by the terms "Contractor", "Authority's Engineer" and "Agreement" respectively.

Notwithstanding anything to the contrary contained in Paragraph 1 above, manual the following Specifications and Standards shall apply to the Project Highway, and for purposes of this Agreement, the aforesaid Specifications and Standards shall bedeemed to be amended to the extent set forth below:

Sl. No.	Clause referred in the Manual	Item	Provision as per Manual	Deviation
Nil				

IRC Codes: (Latest revision)

- IRC: 5-2015 Standard Specifications and Code of Practice for Road Bridges, Section I - General Features of Design
- IRC: 6-2017 Standard Specifications and Code of Practice for Road Bridges, Section-II Loads and Combinations
- IRC: 38 - Guidelines for Design of Horizontal Curves for Highways and Design Tables
- IRC:58-2015 Guidelines for the Design of Plain Jointed Rigid Pavements for Highways (Fourth Revision)
- IRC: 83-2015 (Part I) Standard Specifications and Code of Practice for Road Bridges, Section IX-Bearings, Part I : Roller & Rocker Bearings
- IRC: 83-2018 (Part II) Standard Specifications and Code of Practice for Road Bridges, Section IX – Bearings (Elastomeric Bearings), Part II
- IRC: 83-2018 (Part III) Standard Specifications and Code of Practice for Road Bridges, Section IX – Bearings, Part III: POT-CUM-PTFE, PIN and Metallic Guide Bearings
- IRC: SP:23 - Vertical Curves for Highways
- IRC: SP:42 - Guidelines on Road Drainage
- IRC: SP: 84-2019 Manual of specifications &Standards for Four Laning of Highways with Paved Shoulders
- IS: 6006-1983 (Reaffirmed 2008) – Specification for Uncoated Stress Relieved Strand for Pre-Stressed Concrete

The above latest version of IRC Codes and all other relevant latest IRC applicable as on 28 days prior to bid due date to be considered.

SCHEDULE - E

(See Clauses 2.1 and 14.2)

E. MAINTENANCE REQUIREMENTS

Only Defect Liability Period
Maintenance - Nil

1. Maintenance Requirements

1.1 The Contractor shall, at all times maintain the Project Highway in accordance with the provisions of this Agreement, Applicable Laws and Applicable Permits.

1.2 The Contractor shall repair or rectify any Defect or deficiency set forth in Paragraph 2 of this Schedule-E within the time limit specified therein and any failure in this behalf shall constitute non-fulfillment of the Maintenance obligations by the Contractor. Upon occurrence of any breach hereunder, the Authority shall be entitled to effect reduction in monthly lump sum payment as set forth in Clause 14.6 of this Agreement, without prejudice to the rights of the Authority under this Agreement, including Termination thereof.

1.3 All Materials, works and construction operations shall conform to the MORTH Specifications for Road and Bridge Works, and the relevant IRC publications. Where the specifications for a work are not given, Good Industry Practice shall be adopted.

2. Repair/rectification of Defects and deficiencies

The obligations of the Contractor in respect of Maintenance Requirements shall include repair and rectification of the Defects and deficiencies specified in Annex- I of this Schedule-E within the time limit set forth therein.

3. Other Defects and deficiencies

In respect of any Defector deficiency not specified in Annex-I of this Schedule-E, the Authority's Engineer may, inconformity with Good Industry Practice, specify the permissible limit of deviation or deterioration with reference to the Specifications and Standards, and any deviation or deterioration beyond the permissible limit shall be repaired or rectified by the Contractor within the time limit specified by the Authority's Engineer.

4. Extension of time limit

Not with standing anything to the contrary specified in this Schedule-E, if the nature and extent of any Defector deficiency justifies more time for its repair or rectification than the time specified herein, the Contractor shall be entitled to additional time in conformity with Good Industry Practice. Such

additional time shall be determined by the Authority's Engineer and conveyed to the Contractor and the Authority with reasons thereof.

5. Emergency repairs/restoration

Not with standing anything to the contrary contained in this Schedule-E, if any Defect, deficiency or deterioration in the Project High way poses a hazard to safety or risk of damage to property, the Contractor shall promptly take all reasonable measures for eliminating or minimizing such danger.

6. Daily inspection by the Contractor

The Contractor shall, through its engineer, undertake a daily visual inspection of the Project Highway and maintain are cord thereof in a register to be kept in such form and manner as the Authority's Engineer may specify. Such record shall be kept in safe custody of the Contractor and shall be open to inspection by the Authority and the Authority's Engineer at any time during office hours.

7. Pre-monsoon inspection / Post-monsoon inspection

The Contractor shall carry out a detailed pre-monsoon inspection of all bridges, culverts and drainage system before [1stJune] every year in accordance with the guidelines contained in IRC: SP35. Report of this inspection together with details of proposed maintenance works as required on the basis of this inspection shall be sent to the Authority's Engineer before the [10thJune] every year. The Contractor shall complete the required repairs before the onset of the monsoon and send to the Authority's Engineer a compliance report. Post monsoon inspection shall be done by the [30thSeptember] and the inspection report to get here with details of any damages observed and proposed action to remedy the same shall be sent to the Authority's Engineer.

8. Repairs on account of natural calamities

All damages occurring to the Project Highway on account of a Force Majeure Event or default or neglect of the Authority shall be under taken by the Authority at its own cost. The Authority may instruct the Contractor to undertake the repairs at the rates agreed between the Parties.

Annex – I: Repair/rectification of Defects and deficiencies

(Schedule-E)

The Contractor shall repair and rectify the Defects and deficiencies specified in this Annex-I of Schedule-E within the time limit set forth in the table below.

Table 1: Maintenance Criteria for Pavements:

Asset Type	Performance Parameter	Level of Service (LOS)		Frequency of Inspection	Tools/Equipment	Standards and References for Inspection and Data Analysis	Time limit for Rectification/Repair	Maintenance Specifications
		Desirable	Acceptable					
Flexible Pavement (Pavement of MCW, Service Road, approaches of Grade structure, approaches of connecting roads, slip roads, lay byes etc. as applicable)	Potholes	Nil	< 0.1 % of area and subject to limit of 10 mm in depth	Daily	Length Measurement Unit like Scale, Tape, odometer etc.	IRC 82: 2015 and Distress Identification Manual for Long Term Pavement Performance Program, FHWA 2003 (http://www.tfhrc.com/pavement/lttp/reports/03031/)	24-48 hours	MORT&H Specification 3004.2
	Cracking	Nil	< 5 % subject to limit of 0.5 sqm for any 50 m length	Daily			7-15 days	MORT&H Specification 3004.3
	Rutting	Nil	< 5 mm	Daily	Straight Edge		15 -30 days	MORT&H Specification 3004.2
	Corrugations and	Nil	< 0.1 % of area	Daily	Length Measurement		2-7 days	IRC:82-2015

Asset Type	Performance Parameter	Level of Service (LOS)		Frequency of Inspection	Tools/Equipment	Standards and References for Inspection and Data Analysis	Time limit for Rectification/Repair	Maintenance Specifications
		Desirable	Acceptable					
	Shoving				Unit like Scale, Tape, odometer etc.			
	Bleeding	Nil	< 1 % of area	Daily			3-7 days	MORT&H Specification 3004.4
	Ravelling/ Stripping	Nil	< 1 % of area	Daily			7-15 days	IRC:82-2015 read with IRC SP 81
	Edge Deformation/ Breaking	Nil	< 1 m for any 100 m section and width < 0.1 m at any location, restricted to 30 cm from the edge	Daily			7- 15 days	IRC:82-2015
	Roughness BI	2000 mm/km	2400 mm/km	Bi-Annually	Class I Profilometer SCRIM	Class I Profilometer : ASTM E950 (98) :2004 –Standard Test Method for measuring Longitudinal Profile of Travelled Surfaces with Accelerometer Established Inertial	180 days	IRC:82-2015

Asset Type	Performance Parameter	Level of Service (LOS)		Frequency of Inspection	Tools/Equipment	Standards and References for Inspection and Data Analysis	Time limit for Rectification/Repair	Maintenance Specifications
		Desirable	Acceptable					
	Skid Number	60SN	50SN	Bi-Annually	(Sideway-force Coefficient Routine Investigation Machine or equivalent)	Profiling Reference ASTM E1656 -94: 2000- Standard Guide for Classification of Automatic Pavement Condition Survey Equipment	180 days	BS: 7941-1: 2006
	Pavement Condition Index	3	2.1	Bi-Annually			180 days	IRC:82-2015
	Other Pavement Distresses			Bi-Annually			2-7 days	IRC:82-2015
	Deflection/ Remaining Life			Annually	Falling Weight Deflectometer	IRC 115: 2014	180 days	IRC:115-2014
Rigid Pavement (Pavement of MCW, Service Road, Grade structure, approaches of connecting	Roughness BI	2200mm/km	2400mm/km	Bi-Annually	Class I Profilometer	ASTM E950 (98) :2004 and ASTM E1656 - 94: 2000	180 days	IRC:SP:83-2008
	Skid	Skid Resistance no. at different speed of vehicles **Minimum SN**: 36 **Traffic Speed (Km/h)**: 50		Bi-Annually	SCRIM (Sideway-force Coefficient Routine Investigation Machine or equivalent)	IRC:SP:83-2008	180 days	IRC:SP:83-2008

Asset Type	Performance Parameter	Level of Service (LOS)		Frequency of Inspection	Tools/Equipment	Standards and References for Inspection and Data Analysis	Time limit for Rectification/Repair	Maintenance Specifications
		Desirable	Acceptable					
roads, slip roads, lay byes etc. as applicable)		33	65					
		32	80					
		31	95					
		31	110					
Embankment/ Slope	Edge drop at shoulders	Nil	40mm	Daily	Length Measurement Unit like Scale, Tape, odometer etc.	IRC	7-15 days	MORT&H Specification 408.4
	Slope of camber/cross fall	Nil	<2% variation in prescribed slope of camber /cross fall	Daily			7-15 days	MORT&H Specification 408.4
	Embankment Slopes	Nil	<15 % variation in prescribe side slope	Daily			7-15 days	MORT&H Specification 408.4
	Embankment Protection	Nil	Nil	Daily	NA		7-15 days	MORT&H Specification

Asset Type	Performance Parameter	Level of Service (LOS)		Frequency of Inspection	Tools/Equipment	Standards and References for Inspection and Data Analysis	Time limit for Rectification/Repair	Maintenance Specifications
		Desirable	Acceptable					
	Rain Cuts/ Gullies in slope	Nil	Nil	Daily Specially During Rainy Season	NA		7-15 days	MORT&H Specification

In addition to the above performance criterion, the contractor shall strictly maintain the rigid pavements as per requirements in the following table.

Table 2: Maintenance Criteria for Rigid Pavements:

S.No.	Type of Distress	Measured Parameter	Degree of Severity	Assessment Rating	Repair Action	
					For the case d < D/2	For the case d > D/2
CRACKING						
1	**Single Discrete Cracks Not intersecting with any joint**	w = width of crack L = length of crack d = depth of crack D = depth of slab	0	Nil, not discernible	No Action	Not applicable
			1	w < 0.2 mm. hair cracks		
			2	w = 0.2 - 0.5 mm, discernible from slow-moving car	Seal without delay	Seal, and stitch if L >lm. Within 7days
			3	w = 0.5 - 1.5 mm, discernible from fast-moving car		
			4	w = 1.5 - 3.0 mm	Seal, and stitch if L > l m.	Staple or Dowel Bar

S.No.	Type of Distress	Measured Parameter	Degree of Severity	Assessment Rating	Repair Action	
					For the case d < D/2	**For the case d > D/2**
			5	w > 3 mm.	Within 7 days	Retrofit, FDR for affected portion. Within 15days
2	**Single Transverse (or Diagonal) Crack intersecting with one or more joints**	w = width of crack L = length of crack d = depth of crack D = depth of slab	0	Nil, not discernible	No Action	
			1	w < 0.2 mm, hair cracks	Route and seal with epoxy. Within 7 days	Staple or Dowel Bar Retrofit. Within 15days
			2	w = 0.2 - 0.5 mm, discernible from slow vehicle		
			3	w = 0.5 - 3.0 mm, discernible from fast vehicle	Route, seal and stitch, if L > 1 m. Within 7 days	
			4	w = 3.0 - 6.0 mm	Dowel Bar Retrofit. Within 15 days	Full Depth Repair Dismantle and reconstruct affected. Portion with norms and specifications - See Para 5.5 & 9.2 Within 15days
			5	w > 6 mm, usually associated with spalling, and/or slab rocking under traffic	Not Applicable, as it may be full depth	
3	**Single Longitudinal Crack intersecting with one or more joints**	w = width of crack L = length of crack d = depth of crack D = depth of slab	0	Nil, not discernible	No Action	
			1	w < 0.5 mm, discernable from slow moving vehicle	Seal with epoxy, if L > 1 m. Within 7 days	Staple or dowel bar retrofit. Within 15days

S.No.	Type of Distress	Measured Parameter	Degree of Severity	Assessment Rating	Repair Action	
					For the case d < D/2	**For the case d > D/2**
			2	w = 0.5 - 3.0 mm, discernible from fast vehicle	Route seal and stitch, if L >l m. Within 15 days	-
			3	w = 3.0 - 6.0 mm	Staple, if L > 1 m. Within 15 days	Partial Depth Repair with stapling. Within 15 days
			4	w = 6.0 - 12.0 mm, usually associated with spalling	Not Applicable, as it may be full depth	
			5	w > 12 mm, usually associated with spalling, and/or slab rocking under traffic		Full Depth Repair Dismantle and reconstruct affected portion as per norms and specifications - See Para 5.6.4 Within 15 days
4	**Multiple Cracks intersecting with one or more joints**	**w = width of crack**	0	Nil, not discernible	No Action	-
			1	w < 0.2 mm, hair cracks	If L (Cumulative) < 3 m Full depth repair within 15 Days.	
			2	w = 0.2 - 0.5 mm. discernible from slow vehicle		
			3	w = 0.5 - 3.0 mm, discernible from fast vehicle	If L (Cumulative) > 3 m Dismantle, Reinstate sub-	Dismantle, Reinstate subbase, Reconstruct whole slab as per specifications within 30
			4	w = 3.0 - 6.0 mm panel broken into 2 or 3 pieces		

S.No.	Type of Distress	Measured Parameter	Degree of Severity	Assessment Rating	Repair Action	
					For the case d < D/2	**For the case d > D/2**
			5	w > 6 mm and/or panel broken into more than 4 pieces	base, Reconstruct whole slab as per specifications within 30 days	days
5	**Corner Break**	w = width of crack L = length of crack	0	Nil, not discernible	No Action	-
			1	w < 0.5 mm; only 1 corner broken	Seal with low viscosity epoxy to secure broken parts Within 7 days	Seal with epoxy seal with epoxy Within 7days
			2	w < 1.5 mm; L < 0.6 m, only one corner broken		
			3	w < 1.5 mm; L < 0.6 m, two corners broken	Partial Depth (Refer Figure 8.3 of IRC:SP: 83-2008) Within 15 days	Full depth repair
			4	w > 1.5 mm; L > 0.6 m or three corners broken		
			5	three or four corners broken		Reinstate sub-base, and reconstruct the slab as per norms and specifications within 30days
6	**Punchout (Applicable to Continuous Reinforced Concrete Pavement (CRCP) only)**	w = width of crack L = length (m/m2)	0	Nil, not discernible		No Action
			1	w < 0.5 mm; L < 3 m/m^2	Not Applicable, as it may be full depth	Seal with low viscosity epoxy to secure broken parts. Within 15days
			2	either w > 0.5 mm or L < 3 m/m^2		
			3	w > 1.5 mm and L < 3 m/m^2		
			4	w > 3 mm, L < 3 m/m^2 and deformation		Full depth repair - Cut

S.No.	Type of Distress	Measured Parameter	Degree of Severity	Assessment Rating	Repair Action	
					For the case d < D/2	**For the case d > D/2**
			5	w > 3 mm, L > 3 m/m^2 and deformation		out and replace damaged area taking care not to damage reinforcement. Within 30days
Surface Defects						
7	Ravelling or Honeycomb type surface	r = area damaged surface/total surface of slab (%) h = maximum depth of damage	0	Nil, not discernible	**Short Term**	**Long Term**
					No action.	Not Applicable
			1	r < 2 %	Local repair of areas damaged and liable to be damaged. Within 15 days	
			2	r = 2 - 10 %		
			3	r = 10-25%	Bonded Inlay, 2 or 3 slabs if affecting. Within 30 days	
			4	r = 25 - 50 %		
			5	r > 50% and h > 25 mm	Reconstruct slabs, 4 or more slabs if affecting. Within 30 days	
8	**Scaling**	r = damaged surface/total surface of slab (%) h = maximum depth of damage	0	Nil, not discernible	**Short Term**	**Long Term**
					No action.	Not Applicable
			1	r <**2 %**	Local repair of areas damaged	
			2	r = 2 - 10 %		

S.No.	Type of Distress	Measured Parameter	Degree of Severity	Assessment Rating	Repair Action	
					For the case d < D/2	For the case d > D/2
					and liable to be damaged. Within 7days	
			3	r = 10 - 20%	Bonded Inlay within 15 days	
			4	r = 20 - 30 %		
			5	r > 30 % and h > 25 mm	Reconstruct slab within 30 days	
9	**Polished Surface/Glazing**	t = texture depth, sand patch test	0		No action.	Not Applicable
			1	t > 1 mm		
			2 '	t = 1 - 0.6 mm	Monitor rate of deterioration	
			3	t = 0.6 - 0.3 mm		
			4	t = 0.3 - 0.1 mm		
			5	t < 0.1 mm	Diamond Grinding if affecting 50% or more slabs in a continuous stretch of minimum 5 km. Within 30 days	
10	**Popout (Small Hole), Pothole Refer Para 8.4**	n = number/m^2 d = diameter h = maximum depth	0	d < 50 mm; h < 25 mm; n < 1 per 5 m^2	No action.	Not Applicable
			1	d = 50 - 100 mm; h < 50 mm; n < 1 per 5 m^2	Partial depth repair 65 mm deep.	

S.No.	Type of Distress	Measured Parameter	Degree of Severity	Assessment Rating	Repair Action	
					For the case d < D/2	For the case d > D/2
			2	d = 50 - 100 mm; h > 50 mm; n < 1 per 5 m^2	Within 15 days	
			3	d = 100 - 300 mm; h < 100 mm n < 1 per 5 m^2	Partial depth repair 110mm i.e.10 mm more than the depth of the hole. Within 30 days	
			4	d = 100 - 300 mm; h > 100 mm; n < 1 per 5 m^2		
			5	d > 300 mm; h > 100 mm: n > 1 per 5 m^2	Full depth repair. Within 30 days	

Joint Defects						
			0	Difficult to discern.	Short Term	Long Term
					No action.	Not Applicable
11	**Joint Seal Defects**	loss or damage L = Length as % total joint length	1	Discernible, L< 25% but of little immediate consequence with regard to ingress of water or trapping incompressible material.	Clean joint, inspect later.	
			3	Notable. L > 25% insufficient protection against ingress of water and trapping incompressible	Clean and reapply sealant in selected locations.	

				material.	Within 7 days	
			5	Severe; w > 3 mm negligible protection against ingress of water and trapping incompressible material.	Clean, widen and reseal the joint. Within 7 days	
12	**Spalling of Joints**	w = width on either side of the joint L = length of spalled portion (as % joint length)	0	Nil, not discernible	No action.	Not Applicable
			1	w < 10 mm	Apply low viscosity epoxy resin/ mortar in cracked portion. Within 7 days	
			2	w = 10 - 20 mm, L < 25%		
			3	w = 20 - 40 mm, L > 25%	Partial Depth Repair. Within 15 days	
			4	w = 40 - 80 mm, L > 25%	30 - 50 mm deep, h = w + 20% of w, within 30 days	
			5	w > 80 mm, and L > 25%	50 - 100 mm deep repair. H = w + 20% of w. Within 30 days	
13	**Faulting (or Stepping) in Cracks or Joints**	f = difference of level	0	not discernible, < 1 mm	No action.	No action.
			1	f < 3 mm		
			2	f = 3 - 6 mm	Determine cause and observe, take action for diamond grinding	Replace the slab as appropriate. Within 30days
			3	f = 6 - 12 mm	Diamond Grinding	

			4	f= 12 - 18 mm	Raise sunken slab.	Replace the slab as appropriate. Within 30days
			5	f> 18 mm	Strengthen subgrade and sub-base by grouting and raising sunken slab	
14	**Blowup or Buckling**	h = vertical displacement from normal profile	0	Nil, not discernible	**Short Term**	**Long Term**
			1	h < 6 mm	No Action	
			2	h = 6 - 12 mm	Install Signs to Warn Traffic within 7 days	
			3	h = 12 - 25 mm		
			4	h > 25 mm	Full Depth Repair. Within 30 days	
			5	shattered slabs, ie 4 or more pieces	Replace broken slabs. Within 30 days	
15	**Depression**	h = negative vertical displacement from normal profile L =length	0	Not discernible, h < 5 mm	No action.	Not Applicable
			1	h = 5 - 15 mm		
			2	h = 15-30 mm, Nos <20% joints	Install Signs to Warn Traffic within 7 days	
			3	h = 30 - 50 mm		

			4	h > 50 mm or > 20% joints	Strengthen subgrade. Reinstate pavement at normal level if L < 20 m. Within 30 days	
			5	h > 100 mm		
16	**Heave**	h = positive vertical displacement from normal profile. L = length	0	Not discernible. h < 5 mm	**Short Term**	**Long Term**
					No action.	Scrabble
			1	h = 5 - 15 mm	Follow up.	
			2	h = 15 - 30 mm, Nos <20% joints	Install Signs to Warn Traffic within 7 days	
			3	h = 30 - 50 mm		
			4	h > 50 mm or > 20% joints	Stabilise subgrade. Reinstate pavement at normal level if length < 20 m. Within 30 days	
			5	h > 100 mm		
17	**Bump**	h = vertical displacement from normal profile	0	h < 4 mm	No action	
			1	h = 4 - 7 mm	Grind, in case of new construction within 7 days	Construction Limit for New Construction.
			3	h = 7 - 15 mm	Grind, in case of ongoing Maintenance within 15 days	Replace in case of new construction. Within 30days
			5	h > 15 mm	Full Depth Repair.	Full Depth Repair.

					Within 30 days	Within 30days
18	**Lane to Shoulder Dropoff**	f = difference of level	0	Nil, not discernible < 3mm	**Short Term**	**Long Term**
					No action.	
			1	f = 3 - 10 mm	Spot repair of shoulder within 7 days	
			2	f = 10 - 25 mm		
			3	f = 25 - 50 mm	Fill up shoulder within 7 dayss	
			4	f = 50 - 75 mm		For any 100 m stretch Reconstruct shoulder, if affecting 25% or more of stretch. Within 30days
			5	f > 75 mm		
Drainage						
19	**Pumping**	quantity of fines and water expelled through open joints and cracks Nos	0	not discernible	No Action	
			1 to 2	slight/ occasional Nos < 10%	Repair cracks and joints Without delay.	Inspect and repair sub-drainage at distressed sections and upstream.
			3 to 4	appreciable/ Frequent 10 - 25%	Lift or jack slab within 30 days.	
		Nos/100 m stretch	5	abundant, crack development > 25%	Repair distressed pavement sections. Strengthen subgrade and subbase. Replace slab.	

					Within 30 days	
20	**Ponding**	Ponding on slabs due to blockage of drains	0-2	No discernible problem	No action.	
			3 to 4	Blockages observed in drains, but water flowing	Clean drains etc within 7 days, Follow up	Action required to stop water damaging foundation within 30 days.
			5	Ponding, accumulation of water observed	-do-	

Table -3:Maintenance Criteria for Safety Related Items and Other Furniture Items:

<table>
<tr><th>Asset Type</th><th>Performance Parameter</th><th>Level of Service (LOS)</th><th>Frequency of Measurement</th><th>Testing Method</th><th>Recommended Remedial measures</th><th>Time limit for Rectification</th><th>Specifications and Standards</th></tr>
<tr><td>Highway</td><td>Availability of Safe Sight Distance</td><td>As per IRC SP :84-2014, a minimum of safe stopping sight distance shall be available throughout.
<table>
<tr><th>Design Speed, kmph</th><th>Desirable Minimum Sight Distance (m)</th><th>Safe Stopping Sight Distance (m)</th></tr>
<tr><td>100</td><td>360</td><td>180</td></tr>
<tr><td>80</td><td>260</td><td>130</td></tr>
</table></td><td>Monthly</td><td>Manual Measurements with Odometer along with video/ image backup</td><td colspan="2">Removal of obstruction within 24 hours, in case of sight line affected by temporary objects such as trees, temporary encroachments.
In case of permanent structure or design deficiency:
Removal of obstruction/improvement of deficiency at the earliest
Speed Restriction boards and suitable traffic calming measures such as transverse bar marking, blinkers, etc. shall be applied during the period of rectification.</td><td>IRC:SP 84-2014</td></tr>
<tr><td rowspan="3">Pavement Marking</td><td>Wear</td><td><70% of marking remaining</td><td>Bi-Annually</td><td>Visual Assessment as per Annexure-F of IRC:35-2015</td><td>Re - painting</td><td>Cat-1 Defect – within 24 hours
Cat-2 Defect - within 2 months</td><td>IRC:35-2015</td></tr>
<tr><td>Day time Visibility</td><td>During expected life Service Time
Cement Road - 130mcd/m^2/lux
Bituminous Road - 100mcd/m^2/lux</td><td>Monthly</td><td>As per Annexure-D of IRC:35-2015</td><td>Re - painting</td><td>Cat-1 Defect – within 24 hours
Cat-2 Defect – within 2 months</td><td>IRC:35-2015</td></tr>
<tr><td>Night Time Visibility</td><td><u>Initial and Minimum Performance for</u></td><td>Bi-Annually</td><td>As per Annexure-</td><td>Re - painting</td><td>Cat-1 Defect –</td><td>IRC:35-2015</td></tr>
</table>

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
		Dry Retro reflectivity during night time: Design Speed | (RL) Retro Reflectivity (mcd/m^2/lux) Initial (7 days) | Minimum Threshold level (TL) & warranty period required up to 2 years Up to 65 | 200 | 80 65 - 100 | 250 | 120 Above 100 | 350 | 150 Initial and Minimum Performance for Night Visibility under wet condition (Retro reflectivity): Initial 7 days Retro reflectivity: 100 mcd/m^2/lux Minimum Threshold Level: 50 mcd/m^2/lux		E of IRC:35-2015		within 24 hours Cat-2 Defect – within 2 months	
	Skid Resistance	Initial and Minimum performance for Skid Resistance: Initial (7days): 55BPN Min. Threshold: 44BPN *Note: shall be considered under urban/city traffic condition encompassing the locations like pedestrian crossings,	Bi-Annually	As per Annexure-G of IRC:35-2015		Within 24 hours	IRC:35-2015

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
		bus bay, bus stop, cycle track intersection delineation, transverse bar markings etc					
Road Signs	Shape and Position	Shape and Position as per IRC:67-2012. Signboard should be clearly visible for the design speed of the section.	Daily	Visual with video/image backup	Improvement of shape, in case if shape is damaged. Relocation as per requirement	48 hours in case of Mandatory Signs, Cautionary and Informatory Signs (Single and Dual post signs) 15 Days in case of Gantry/Cantilever Sign boards	IRC:67-2012
	Retro reflectivity	As per specifications in IRC:67-2012	Bi-Annually	Testing of each signboard using Retro Reflectivity Measuring Device. In accordance with ASTM D 4956-09.	Change of signboard	48 hours in case of Mandatory Signs, Cautionary and Informatory Signs (Single and Dual post signs) 1 Month in case of Gantry/Cantilever Sign boards	RC:67-2012
Kerb	Kerb Height	As per IRC 86:1983 depending upon type of Kerb	Bi-Annually	Use of distance measuring tape	Raising Kerb Height	Within 1 Month	RC 86:1983
	Kerb Painting	Functionality: Functioning of Kerb painting as intended	Daily	Visual with video/image backup	Kerb Repainting	Within 7-days	RC 35:2015
Other Road Furniture	Reflective Pavement Markers (Road Studs)	Numbers and Functionality as per specifications in IRC:SP:84-2014 and IRC:35-2015, unless specified in Schedule-B.	Daily	Counting	New Installation	Within 2 months	IRC:SP:84-2014, IRC:35-2015

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
	Pedestrian Guardrail	Functionality: Functioning of guardrail as intended	Daily	Visual with video/image backup	Rectification	Within 15 days	IRC:SP:84-2014
	Traffic Safety Barriers	Functionality: Functioning of Safety Barriers as intended	Daily	Visual with video/image backup	Rectification	Within 7 days	IRC:SP:84-2014, IRC:119-2015
	End Treatment of Traffic Safety Barriers	Functionality: Functioning of End Treatment as intended	Daily	Visual with video/image backup	Rectification	Within 7 days	IRC:SP:84-2014, IRC:119-2015
	Attenuators	Functionality: Functioning of Attenuators as intended	Daily	Visual with video/image backup	Rectification	Within 7 days	IRC:SP-2014, IRC:119-2015
	Guard Posts and Delineators	Functionality: Functioning of Guard Posts and Delineators as intended	Daily	Visual with video/image backup	Rectification	Within 15 days	IRC: 79 – 1981
	Overhead Sign Structure	Overhead sign structure shall be structurally adequate	Daily	Visual with video/image backup	Rectification	Within 15 days	IRC:67-2012
	Traffic Blinkers	Functionality: Functioning of Traffic Blinkers as intended	Daily	Visual with video/image backup	Rectification	Within 7 days	IRC:SP:84-2014
Highway Lighting System	Highway Lights	Illumination: Minimum 40 Lux illumination on the road surface	Daily	The illumination level shall be measured with luxmeter	Improvement in Lighting System	24 hours	IRC:SP:84-2014
		No major failure in the lighting system	Daily	-	Rectification of failure	24 hours	IRC:SP:84-2014
		No minor failure in the lighting system	Monthly	-	Rectification of failure	8 hours	IRC:SP:84-2014
	Toll Plaza Canopy Lights	Minimum 40 Lux illumination on the road surface	Daily	The illumination level shall be measured with	Improvement in Lighting System	24 hours	IRC:SP:84-2014

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
				luxmeter			
		No major/minor failure in the lighting system	Daily	-	Rectification of failure	8 hours	IRC:SP:84-2014
Trees and Plantation including median plantation	Obstruction in a minimum head-room of 5.5 m above carriageway or obstruction in visibility of road signs	No obstruction due to trees	Monthly	Visual with video/image backup	Removal of trees	Immediate	IRC:SP:84-2014
	Deterioration in health of trees and bushes	Health of plantation shall be as per requirement of specifications & instructions issued by Authority from time to time	Daily	Visual with video/image backup	Timely watering and treatment. Or Replacement of Trees and Bushes.	Within 90 days	IRC:SP:84-2014
	Vegetation affecting sight line and road structures	Sight line shall be free from obstruction by vegetation	Daily	Visual with video/image backup	Removal of Trees	Immediate	IRC:SP 84-2014
Rest Areas	Cleaning of toilets	-	Daily	-	-	Every 4 hours	
	Defects in electrical, water and sanitary installations	-	Daily	-	Rectification	24 hours	

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
Other Project Facilities and Approach roads	Damage or deterioration in Approach Roads, pedestrian facilities, truck lay-bys, bus-bays, bus- shelters, cattle crossings, Traffic Aid Posts, Medical Aid Posts and other works		Daily	-	Rectification	15 days	IRC:SP 84-2014

Table 1: Maintenance Criteria for Structures and Culverts:

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
Pipe/box/slab culverts	Free waterway/unobstructed flow section	85% of culvert normal flow area to available.	2 times in a year (before and after rainy season)	Inspection by Bridge Engineer as per IRC SP: 35-1990 and recording of depth of silting and area of vegetation.	Cleaning silt up soils and debris in culvert barrel after rainy season, removal of bushes and vegetation, U/s of barrel, under barrel and D/s of barrel before rainy season.	15 days before onset of monsoon and within 30 days after end of rainy season.	IRC 5-2015, IRC SP:40-1993 and IRC SP:13-2004
	Leak-proof expansion joints if any	No leakage through expansion joints	Bi-Annually	Physical inspection of expansion joints as per IRC SP: 35-1990 if any,	Fixing with sealant suitably	30 days or before onset of rains whichever comes earlier	IRC SP:40-1993 and IRC SP:69-2011

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
				for leakage strains on walls at joints.			
	Structurally sound	Spalling of concrete not more than 0.25 sqm	Bi-Annually	Detailed inspection of all components of culvert as per IRC SP:35-1990 and recording the defects	Repairs to spalling, cracking, delamination, rusting shall be followed as per IRC:SP:40-1993.	15 days	IRC SP 40-1993 and MORTH Specifications clause 2800
		Delamination of concrete not more than 0.25 sq.m.					
		Cracks wider than 0.3 mm not more than 1m aggregate length					
	Protection works in good condition	Damaged of rough stone apron or bank revetment not more than 3 sqm, damage to solid apron (concrete apron) not more than 1 sqm	2 times in a year (before and after rainy season)	Condition survey as per IRC SP:35-1990	Repairs to damaged aprons and pitching	30 days after defect observation or 2 weeks before onset of rainy season whichever is earlier.	IRC: SP 40-1993 and IRC:SP:13-2004.
Bridges including ROBs Flyover etc. as applicable	Riding quality or user comfort	No pothole in wearing coat on bridge deck	Daily	Visual inspection as per IRC SP:35-1990	Repairs to BC or wearing coat	15 days	MORT&H Specification 2811
Bridge -Super Structure	Bumps	No bump at expansion joint	Daily	Visual inspection as per IRC SP:35-1990	Repairs to BC on either side of expansion joints, profile correction	15 days	MORT&H Specification 3004.2 & 2811.

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
					course on approach slab in case of settlement to approach embankment		
	User safety (condition of crash barrier and guard rail)	No damaged or missing stretch of crash barrier or pedestrian hand railing	Daily	Visual inspection and detailed condition survey as per IRC SP: 35-1990.	Repairs and replacement of safety barriers as the case may be	3days	IRC: 5-1998, IRC SP: 84-2014 and IRC SP: 40-1993.
	Rusted reinforcement	Not more than 0.25 sq.m	Bi-Annually	Detailed condition survey as per IRC SP: 35-1990 using Mobile Bridge Inspection Unit	All the corroded reinforcement shall need to be thoroughly cleaned from rusting and applied with anti-corrosive coating before carrying out the repairs to affected concrete portion with epoxy mortar / concrete.	15 days	IRC SP: 40-1993 and MORTH Specification 1600.
	Spalling of concrete	Not more than 0.50 sq.m					
	Delamination	Not more than 0.50 sq.m					
	Cracks wider than 0.30 mm	Not more than 1m total length	Bi-Annually	Detailed condition survey as per IRC SP: 35-1990 using Mobile Bridge Inspection Unit	Grouting with epoxy mortar, investigating causes for cracks development and carry out necessary	48 Hours	IRC SP: 40-1993 and MORTH Specification 2800.

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
					rehabilitation.		
	Rainwater seepage through deck slab	Leakage - nil	Quarterly	Detailed condition survey as per IRC SP: 35-1990 using Mobile Bridge Inspection Unit	Grouting of deck slab at leakage areas, waterproofing, repairs to drainage spouts	1 months	MORTH specifications 2600 & 2700.
	Deflection due to permanent loads and live loads	Within design limits.	Once in every 10 years for spans more than 40 m	Load test method	Carry out major rehabilitation works on bridge to retain original design loads capacity	6 months	IRC SP: 51-1999.
	Vibrations in bridge deck due to moving trucks	Frequency of vibrations shall not be more than 5 Hz	Once in every 5 years for spans more than 30m and every 10 years for spans between 15 to 30 m	Laser displacement sensors or laser vibro-meters	Strengthening of super structure	4 months	AASHTO LRFD specifications
	Leakage in Expansion joints	No damage to elastomeric sealant compound in strip seal expansion joint, no leakage of rain water through expansion joint in case of buried and asphalt plug and copper strip joint.	Bi-Annually	Detailed condition survey as per IRC SP:35-1990 using Mobile Bridge Inspection Unit	Replace of seal in expansion joint	15 days	MORTH specifications 2600 and IRC SP: 40-1993.
	Debris and dust in strip	No dust or debris in	Monthly	Detailed	Cleaning of	3 days	MORTH

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
	seal expansion joint	expansion joint gap.		condition survey as per IRC SP:35-1990 using Mobile Bridge Inspection Unit	expansion joint gaps thoroughly		specifications 2600 and IRC SP: 40-1993.
	Drainage spouts	No down take pipe missing/broken below soffit of the deck slab. No silt, debris, clogging of drainage spout collection chamber.	Monthly	Detailed condition survey as per IRC SP: 35-1990 using Mobile Bridge Inspection Unit	Cleaning of drainage spouts thoroughly. Replacement of missing/broken down take pipes with a minimum pipe extension of 500mm below soffit of slab. Providing sealant around the drainage spout if any leakages observed.	3 days	MORTH specification 2700.
Bridge-substructure	Cracks/spalling of concrete/rusted steel	No cracks, spalling of concrete and rusted steel	Bi-Annually	Detailed condition survey as per IRC SP: 35-1990 using Mobile Bridge Inspection Unit	All the corroded reinforcement shall need to be thoroughly cleaned from rusting and applied with anti-corrosive coating before carrying out repairs to substructure by	30 days	IRC SP: 40-1993 and MORTH specification 2800.

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
					grouting/guniting and micro concreting depending on type of defect noticed		
	Bearings	Delamination of bearing reinforcement not more than 5%, cracking or tearing of rubber not more than 2 locations per side, no rupture of reinforcement or rubber	Bi-Annually	Detailed condition survey as per IRC SP: 35-1990 using Mobile Bridge Inspection Unit	In case of failure of even one bearing on any pier/abutment, all the bearings on that pier/abutment shall be replaced, in order to get uniform load transfer on to bearings.	3 months	MORTH specification 2810 and IRC SP: 40-199.
Bridge Foundations	Scouring around foundations	Scouring shall not be lower than maximum scour level for the bridge	Bi-Annually	Condition survey and visual inspection as per IRC SP:35-1990 using Mobile Bridge Inspection Unit. In case of doubt, use Underwater camera for inspection of deep wells in	Suitable protection works around pier/abutment	1 month	IRC SP: 40-1993, IRC 83-2014, MORTH specification 2500

Asset Type	Performance Parameter	Level of Service (LOS)	Frequency of Measurement	Testing Method	Recommended Remedial measures	Time limit for Rectification	Specifications and Standards
				major Rivers.			
	Protection works in good condition	Damaged of rough stone apron or bank revetment not more than 3 sq.m, damage to solid apron (concrete apron) not more than 1 sq.m	2 times in a year (before and after rainy season)	Condition survey as per IRC SP:35-1990	Repairs to damaged aprons and pitching.	30 days after defect observation or 2 weeks before onset of rainy season whichever is earlier.	IRC: SP 40-1993 and IRC:SP:13-2004.
Note: Any Structure during the entire contract period which is found that does not complies with all requirements of this Table will be prepared, rehabilitated or even reconstructed under the scope of the contractor.							

Table 5: Maintenance Criteria for Hill Roads

In addition to above, for hill roads the following provisions for maintenance is also to done.

Hill Roads		
(i)	Damage to Retaining wall/ Breast wall	7 (Seven) days
(ii)	Landslides requiring clearance	12 (Twelve) hours
(iii)	Snow requiring clearance	24 (Twenty Four) hours

Note: For all tables 1 to 5 above, latest BIS & IRC standards (even those not indicated herewith) along with MoRTH specifications shall be binding for all maintenance activities.

A. Flexible Pavement

Nature of Defect or deficiency	Time limit for repair/ rectification
(b) Granular earth shoulders, side slopes, drains and culverts	

	Nature of Defect or deficiency	Time limit for repair/ rectification
(i)	Variation by more than 1 % inthe prescribed slope of camber/cross fall (shall not be less than the camber on the maincarriageway)	7 (seven) days
(ii)	Edge drop at shoulders exceeding 40 mm	7 (seven) days
(iii)	Variationbymorethan15%inthe prescribed side (embankment) slopes	30 (thirty) days
(iv)	Rain cuts/gullies in slope	7 (seven) days
(v)	Damage to or silting of culverts and side drains	7 (seven) days
(vi)	Desilting of drains in urban/semi- urban areas	24 (twenty four) hours
(vii)	Railing, parapets, crash barriers	7 (seven) days (Restore immediately if causing safety hazard)
(c)	**Road side furniture including road sign and pavement marking**	
(i)	Damagetoshapeorposition,poor visibility or loss of retro- reflectivity	48 (forty eight) hours
(ii)	Painting of km stone, railing, parapets, crash barriers	As and when required/ Once everyyear
(iii)	Damaged/missing signs road requiringreplacement	7 (seven) days
(iv)	Damage to road mark ups	7 (seven) days
(d)	**Road lighting**	
(i)	Any major failure of the system	24 (twenty four) hours
(ii)	Faults and minor failures	8 (eight) hours
(e)	**Trees and plantation**	
(i)	Obstruction in a minimum head- room of 5 m above carriagewayor obstruction in visibility of road signs	24 (twenty four)hours
(ii)	Removal of fallen trees from carriageway	4 (four) hours
(iii)	Deterioration in health of trees and bushes	Timely watering and treatment
(iv)	Trees and bushes requiring replacement	30 (thirty) days
(v)	Removal of vegetation affecting sight line and road structures	15 (fifteen) days
(f)	**Rest area**	

Nature of Defect or deficiency		Time limit for repair/ rectification
(i)	Cleaning of toilets	Every 4 (four) hours
(ii)	Defects in electrical, water and sanitary installations	24 (twenty four) hours
(g)	**[Toll Plaza]**	
(h)	**Other Project Facilities and Approachroads**	
(i)	Damage in approach roads, pedestrian facilities, truck lay- byes, bus-bays, bus-shelters,cattle crossings, [Traffic Aid Posts, Medical Aid Posts] and service roads	15 (fifteen) days
(ii)	Damaged vehicles or debris on the road	4 (four) hours
(iii)	Malfunctioning of the mobile crane	4 (four) hours
Bridges		
(a)	**Superstructure**	
(i)	Any damage, cracks, sapling/ scaling Temporary measures Permanent measures	within 48 (forty eight) hours within15(fifteen)daysor as specified by the Authority'sEngineer
(b)	**Foundations**	
(i)	Scouring and/or cavitations	15 (fifteen) days
(c)	**Piers, abutments, return walls and wing walls**	
(i)	Cracks and damagesincluding settlement and tilting, spalling, scaling	30 (thirty) days
(d)	**Bearings (metallic) of bridges**	
(i)	Deformation, damages, tilting or shifting of bearings	15 (fifteen) days Greasing of metallic bearings once in a year
(e)	**Joints**	
(i)	Malfunctioning of joints	15 (fifteen) days
(f)	**Other items**	
(i)	Deforming of pads in elastomeric bearings	7 (seven) days
(ii)	Gathering of dirt in bearings and joints; or clogging of spouts,weep holes andvent-holes	3 (three) days

Nature of Defect or deficiency		Time limit for repair/ rectification
(iii)	Damage or deterioration in kerbs, parapets, handrails andcrash barriers	3 (three) days (immediately within 24 hours if posing danger to safety)
(iv)	Rain-cuts or erosion of banks of the side slopes of approaches	7 (seven) days
(v)	Damage to wearing coat	15 (fifteen) days
(vi)	Damage or deterioration in approach slabs, pitching, apron, toes, floor or guide bunds	30 (thirty) days
(vii)	Growth of vegetation affectingthe structure or obstructingthe waterway	15 (fifteen) days
(g)	**Hill Roads**	
(i)	Damage to retaining wall/breast wall	7 (seven) days
(ii)	Landslides requiring clearance	12 (twelve) hours
(iii)	Snow requiring clearance	24 (twenty four) hours

[Note: Where necessary, the Authority may modify the time limit for repair/rectification, or add to the nature of Defect or deficiency before issuing the bidding document, with the approval of the competent authority.]

SCHEDULE - F

(See Clause 3.1.7(a))

F. APPLICABLE PERMITS

1. Applicable Permits

1.1 The Contractor shall obtain, as required under the Applicable Laws, the following Applicable Permits:

(a) Permission of the State Government for extraction of boulders from quarry;

(b) Permission of Village Panchayats and Pollution Control Board for installation of crushers;

(c) License for use of explosives;

(d) Permission of the State Government for drawing water from river/reservoir;

(e) License from inspector off factories or other competent Authority for setting up batching plant;

(f) Clearance of Pollution Control Board for setting up batching plant;

(g) Clearance of Village Panchayats and Pollution Control Board for setting up asphalt plant;

(h) Permission of Village Panchayats and State Government for borrow earth; and

(i) Any other permits or clearances required under Applicable Laws.

1.2 Applicable Permits, as required, relating to environmental protection and conservation shall have been procured by the Authority in accordance with the provisions of this Agreement.

SCHEDULE - G

(See Clauses 7.1 and 19.2)

G. FORM OF BANK GUARANTEE

Annex-I Performance Security/Additional Performance Security

(See Clause 7.1.1)

Chief Engineer, National Highways (P.W.),
Government of Maharashtra
Room No. 526; 5th Floor, Konkan Bhawan
Sector –10, CBD Belapur
Navi Mumbai – 400 614

WHEREAS:

(A) [name and address of contractor] (hereinafter called the "**Contractor**")and[Chief Engineer, National Highways (P.W.), Government of Maharashtra, Room No. 526; 5th Floor, Konkan Bhawan, Sector –10, CBD Belapur, Navi Mumbai – 400 614],(hereinafter called the **"Authority"**) have entered into an agreement (hereinafter called the "**Agreement**") for the Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra on EPC Mode, subject to and in accordance with the provisions of the Agreement

(B) The Agreement requires the Contractor to furnish a Performance Security for due and faithful performance of its obligations, under and in accordance with the Agreement, during the {Construction Period/ Defects Liability Period and Maintenance Period} (as defined in the Agreement) in a sum of Rs….. cr. (Rupees ……..….. crore) (the "**Guarantee Amount**").

(C) We,……..through our branch at…….(the "**Bank**") have agreed to furnish this bank guarantee (*hereinafter called the* **"Guarantee**") by way of Performance Security.

NOW, THEREFORE, the Bank hereby, unconditionally and irrevocably, guarantees and affirms as follows:

1. The Bank hereby unconditionally and irrevocably guarantees the due and faithful performance of the Contractor's obligations during the {Construction Period/ Defects Liability Period and Maintenance Period} under and in accordance with the Agreement, and agrees and undertakes to pay to the Authority, upon its mere first written demand, and without any demur, reservation, recourse, contest or protest, and without any reference to the Contractor, such sum or sums up to an aggregate sum of the Guarantee Amount as the Authority shall claim, without the Authority being required to prove or to show grounds or reasons for its demand and/or for the sum specified therein.

2. A letter from the Authority, under the hand of an officer not below the rank of [General Manager in the National Highways Authority of India], that the Contractor has committed default in the due and faithful performance of all or any of its obligations under and in accordance with the Agreement shall be conclusive, final and binding on the Bank. The Bank further agrees that the Authority shall be the

sole judge as to whether the Contractor is in default in due and faithful performance of its obligations during and under the Agreement and its decision that the Contractor is in default shall be final and binding on the Bank, notwithstanding any differences between the Authority and the Contractor, or any dispute between them pending before any court, tribunal, arbitrators or any other authority or body, or by the discharge of the Contractor for any reason whatsoever.

3. In order to give effect to this Guarantee, the Authority shall be entitled to act as if the Bank were the principal debtor and any change in the constitution of the Contractor and/or the Bank, whether by their absorption with any other body or corporation or otherwise, shall not in any way or manner affect the liability or obligation of the Bank under this Guarantee.

4. It shall not be necessary, and the Bank hereby waives any necessity, for the Authority to proceed against the Contractor before presenting to the Bank its demand under this Guarantee.

5. The Authority shall have the liberty, without affecting in any manner the liability of the Bank under this Guarantee, to vary at any time, the terms and conditions of the Agreement or to extend the time or period for the compliance with, fulfilment and/ or performance of all or any of the obligations of the Contractor contained in the Agreement or to postpone for any time, and from time to time, any of the rights and powers exercise able by the Authority against the Contractor, and either to enforce or for bear from enforcing any of the terms and conditions contained in the Agreement and/or the securities available to the Authority, and the Bank shall not be released from its liability and obligationunderthesepresentsbyanyexercisebytheAuthorityoftheliberty with reference to the matters aforesaid or by reason of time being given to the Contractor or any other forbearance, indulgence, act or omission on the part of the Authority or of any other matter or thing whatsoever which under any law relating to sureties and guarantors would but for this provision have the effect of releasing the Bank from its liability and obligation under this Guarantee and the Bank hereby waives all of its rights under any such law.

6. This Guarantee is in addition to and not in substitution of any other guarantee or security now or which may hereafter be held by the Authority in respect of or relating to the Agreement or for the fulfilment, compliance and/or performance of all or any of the obligations of the Contractor under the Agreement.

7. Notwithstanding anything contained hereinbefore, the liability of the Bank underthisGuaranteeisrestrictedtotheGuaranteeAmountandthisGuarantee will remain in force for the period specified in paragraph 8 below and unless a demand or claim in writing is made by the Authority on the Bank under this Guarantee all rights of the Authority under this Guarantee shall be forfeited and the Bank shall be relieved from its liabilities hereunder.

8. The Guarantee shall cease to be in force and effect on****[$].Unlessademand or claim under this Guarantee is made in writing before expiry of the Guarantee, the Bank shall be discharged from its liabilities hereunder.

[$] Insert date being 2 (two) years from the date of issuance of this Guarantee (in accordance with Clause 7.2 of the Agreement).

9. The Bank undertakes not to revoke this Guarantee during its currency, except with the previous express consent of the Authority in writing, and declares and warrants that it has the power to issue this Guarantee and the undersigned has full powers to do so on behalf of the Bank.

10. Any notice by way of request, demand or otherwise hereunder may be sent by post addressed to the Bank at its above referred branch, which shall be deemed to have been duly authorised to receive such notice and to effect payment thereof forthwith, and if sent by post it shall be deemed to have been given at the time when it ought to have been delivered in due course of post and in proving such notice, when given by post, it shall be sufficient to prove that the envelope containing the notice was posted and a certificate signed by an officer of the Authority that the envelope was so posted shall be conclusive.

11. This Guarantee shall come into force with immediate effect and shall remain in force and effect for up to the date specified in paragraph 8 above or until it is released earlier by the Authority pursuant to the provisions of the Agreement.

Signed and sealed this ……. day of …….., 20…… at ……..
SIGNED, SEALED AND DELIVERED
For and on behalf of the Bank by: (Signature)
(Name)
(Designation)
(Code Number)
(Address)

NOTES:

(i) The bank guarantee should contain the name, designation and code number of the officer(s) signing the guarantee.

(ii) The address, telephone number and other details of the head office of the Bank as well as of issuing branch should be mentioned on the covering letter of issuing branch.

Annex – II: Form for Guarantee for Advance Payment

(Schedule - G)

(SeeClause19.2)

Chief Engineer, National Highways (P.W.),
Government of Maharashtra
Room No. 526; 5th Floor, Konkan Bhawan
Sector –10, CBD Belapur
Navi Mumbai – 400 614

WHEREAS:

(A) [name and address of contractor] (hereinafter called the "**Contractor**") has executed an agreement (hereinafter called the "**Agreement**") with the [Chief Engineer, National Highways (P.W.), Government of Maharashtra, Room No. 526; 5th Floor, Konkan Bhawan, Sector –10, CBD Belapur, Navi Mumbai – 400 614], (hereinafter called the "**Authority**") for the Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra on EPC Mode, subject to and in accordance with the provisions of the Agreement

(B) InaccordancewithClause19.2oftheAgreement,theAuthorityshallmaketo the Contractor an interest bearing *@Bank Rate + 3%* advance payment(herein after called "**Advance Payment**") equal to 10% (tenpercent) of the Contract Price; and that the Advance Payment shall be made in two instalments subject to the Contractor furnishing an irrevocable and unconditional guarantee by a scheduled bank for an amount equivalent to 110% (one hundred and ten percent) of such instalment to remain effective tillthecompleteandfullrepaymentoftheinstallmentoftheAdvancePayment as security for compliance with its obligations in accordance with the Agreement. The amount of{first/second}instalment of the Advance Payment is Rs. ------ cr. (Rupees ------ crore) and the amount of this Guarantee is Rs.------- cr. (Rupees ------ crore) (the **"Guarantee Amount")[$].**

(C) We,……..through our branch at…….(the "**Bank"**) have agreed to furnish this bank guarantee (*hereinafter called the* **"Guarantee"**) for the Guarantee Amount.

NOW, THEREFORE, the Bank hereby, unconditionally and irrevocably, guarantees and affirms as follows:

1. The Bank hereby unconditionally and irrevocably guarantees the due and faithfulrepaymentontimeoftheaforesaidinstalmenttotheAdvancePayment under and in accordance with the Agreement, and agrees and undertakes to pay to the Authority ,upon itsmere first written demand, and without any demur, reservation, recourse, contest or protest, and without any reference to the Contractor, such sum or sums up to an aggregate sum of the Guarantee Amount as the Authority shall claim, without the Authority being required to prove or to show grounds or reasons for its demand

[$] The Guarantee Amount should be equivalent to 110% of the value of the applicable instalment.

and/or for the sum specified therein.

A letter from the Authority, under the hand of an officer not below the rank of [General Manager in the National Highways Authority of India], that the Contractor has committed default in the due and faithful performance of all or any of its obligations for the repayment of the instalment of the Advance Payment under and in accordance with the Agreement shall be conclusive, final and binding on the Bank. The Bank further agrees that the Authority shall be the sole judge as to whether the Contractor is in default in due and faithful performance of its obligations during and under the Agreement and its decision that the Contractor is in default shall be final and binding on the Bank, not withstanding any differences between the Authority and the Contractor, or any dispute between them pending before any court, tribunal, arbitrators or any other authority or body, or by the discharge of the Contractor for any reason what so ever.

2. In order to give effect to this Guarantee, the Authority shall be entitled to act as if the Bank were the principal debtor and any change in the constitution of the Contractor and/or the Bank, whether by their absorption with any other body or corporation or otherwise, shall not in any way or manner affect the liability or obligation of the Bank under this Guarantee.

3. It shall not be necessary, and the Bank hereby waives any necessity, for the Authority to proceed against the Contractor before presenting to the Bank its demand under this Guarantee.

4. The Authority shall have the liberty, without affecting in any manner the liability of the Bank under this Guarantee, to vary at any time, the terms and conditions of the Advance Payment or to extend the time or period of its repaymentortopostponeforanytime,andfromtimetotime,anyoftherights and powers exercisable by the Authority against the Contractor, and either to enforce or forbear from enforcing any of the terms and conditions contained in the Agreement and/or the securities available to the Authority, and the Bank shall not be released from its liability and obligation under these presents by any exercise by the Authority of the liberty with reference to the matters aforesaid or by reason of time being given to the Contractor or any other forbearance, indulgence, act or omission on the part of the Authority or of any other matter or thing whatsoever which under any law relating to sureties and guarantors would but for this provision have the effect of releasing the Bank from its liability and obligation under this Guarantee and the Bank hereby waives all of its rights under any such law.

5. This Guarantee is in addition to and not in substitution of any other guarantee or security now or which may hereafter be held by the Authority in respect of or relating to the Advance Payment.

6. Notwithstanding anything contained hereinbefore, the liability of the Bank under this Guarantee is restricted to the Guarantee Amount and this Guarantee will remain in force for the period specified in paragraph 8 below and unless a demand or claim in writing is made by the Authority on the Bank under this Guarantee all rights of the Authority under this Guarantee shall be forfeited and the Bank shall be relieved from its liabilities here under.

7. The Guarantee shall cease to be inforce and effect on****.[$]Unlessademand or claim under this Guarantee is made in writing on or before the aforesaid date, the Bank shall be discharged from its

[$]Insert a date being 90 (ninety) days after the end of one year from the date of payment of the Advance payment to the Contractor (in accordance with Clause 19.2 of the Agreement).

liabilities here under.

8. The Bank undertakes not to revoke this Guarantee during its currency, except with the previous express consent of the Authority in writing, and declares and warrants that it has the power to issue this Guarantee and the undersigned has full powers to do so on behalf of the Bank.

9. Any notice by way of request, demand or otherwise hereunder may be sent by post addressed to the Bank at its above referred branch, which shall be deemed to have been duly authorised to receive such notice and to effect payment thereof forthwith, and if sent by post it shall be deemed to have been given at the time when it ought to have been delivered in due course of post and in proving such notice, when given by post, it shall be sufficient to prove that the envelope containing the notice was posted and a certificate signed by an officer of the Authority that the envelope was so posted shall be conclusive.

10. This Guarantee shall come into force with immediate effect and shall remain in force and effect up to the date specified in paragraph 8 above or until it is released earlier by the Authority pursuant to the provisions of the Agreement.

Signed and sealed this ……. day of …….., 20…… at ……..

SIGNED, SEALED AND DELIVERED

For and on behalf of the Bank by: (Signature)

(Name)

(Designation)

(Code Number)

(Address)

NOTES:

(i) The bank guarantee should contain the name, designation and code number of the officer(s) signing the guarantee.

(ii) The address, telephone number and other details of the head office of the Bank as well as of issuing branch should be mentioned on the covering letter of issuing branch.

SCHEDULE - H

(See Clauses10.1.4 and 19.3)

H. CONTRACT PRICE WEIGHTAGES

1.1 The Contract Price for this Agreement is **Rs 12,05,33,835/-.**

1.2 Proportions of the Contract Price for different stages of Construction of the Project Highway shall be as specified below:

Item	Weightage in percentage to the Contract Price	Stage for Payment	Percentage Weightage in particular Item
Road works including Widening & Construction of New Culverts & Site Clearance	**42.03%**	**A.1: Reconstruction / New 4 lane / realignment (Rigid Pavement)**	
		Earthwork up to top of the Sub-grade, GSB, DLC & PQC& Granular Shoulder complete.	-
		A.2: Reconstruction / New 2 lane / realignment (Rigid Pavement)	
		Earthwork up to top of the Sub-grade, GSB, DLC & PQC	**93.08%**
		B. Reconstruction of Culverts	**6.43%**
		C. Site Clearance including dismantling and Tree Cutting	**0.49%**
~~Minor Bridges~~	-	~~**Repair & Balance work of Existing Minor Bridges (Length >6m and <60m)**~~	-
		~~**Construction of Balance Protection Work and River Training Works:** On completion of River Training Works & Protection Works (parapet, slope protection, wearing coat), complete in all respects~~	-
Other works (Drains, Junction Development, Paver Blocks, Kerb Street Lights, High Mast Safety Work, etc.)	**39.04%**	**Miscellaneous:**	
		(i) Encased RCC Pipe Drain with Paved Footpath Over It for built up area	**59.93%**
		(ii) Barrier Type Kerb	**1.81%**
		(iii) Road Markings, Road Signs, Solar Blinker, Rumble Strips, Crash Barrier, Studs, Delineators, Safety Boards, Hazard Markers, etc.	**1.66%**
		(iv) Street Lighting & High Mast	**8.54%**
		(v) Junctions Development	**14.24%**
		(vi) Paver Block including Crust under it, complete	**13.08%**
		(vii) Utility Pipe ducts	**0.74%**
Electrical Utilities	**12.41%**	**Electrical Utility Shifting:**	**100%**
		LT U/G Line	
		11 KV U/G Line	
		DP Structure with AB Switch	
		SPCP (for Cable support)	

		Shifting of 200 KVA Transformers	
		Shifting of 100 KVA Transformers	
Water Utilities	**6.52%**	Shifting of CI Pipe	**100%**
		Shifting of HDPE Pipe	

1.3 Procedure of estimating the value of work done:

1.3.1 Road works including approaches to Minor bridges, Major Bridges and Structures (excluding service roads).
Procedure for estimating the value of road work done shall be as follows:

Table 1.3.1

Stage of Payment	Percentage Weightage	Payment Procedure
A.1: Reconstruction / New 4 lane / realignment (Rigid Pavement)		
Earthwork up to top of the sub-grade, GSB, DLC & PQC	-	Unit of measurement is linear length. Payment of each stage shall be made on pro rata basis on completion of a stage in full length in this stretch.
A.2: Reconstruction / New 2 lane / realignment (Rigid Pavement)		
Earthwork up to top of the sub-grade, GSB, DLC & PQC	**93.08%**	Unit of measurement is linear length. Payment of each stage shall be made on pro rata basis on completion of a stage in full length in this stretch.
B. Reconstruction of Culverts (Box & HPC)	**6.43%**	Payment shall be made on the completion of culvert work.
C. Site Clearance including dismantling and Tree Cutting	**0.49%**	Payment shall be made on pro rata basis for completed facilities.

@ For example, if the total length of bituminous work to be done is 100 km, the cost per km of bituminous work shall be determined as follows:

Cost per km = P x weightage for road work x weightage for bituminous work x (1/L)

Where P = Contract Price

L = Total length in km

Similarly, the rates per km for stages shall be worked out accordingly.

Note: The length affected due to law-and-order problems or litigation during execution due to which the contractor is unable to execute the work, may be deducted from the total project length for payment purposes. The total length calculated here is only for payment purposes and will not affect and referred in other clauses of the contract Agreement

1.3.2 Minor Bridges and Underpasses/Overpasses.
Procedure for estimating the value of Minor bridge and Underpasses/Overpasses shall be as stated in table 1.3.2

Table 1.3.2

~~Stage of Payment~~	~~Percentage Weightage~~	~~Payment Procedure~~

~~Repair & Balance work of Existing Minor Bridges (Length >6m and <60m)~~		
~~**Construction of Balance Protection Work and River Training Works:** On completion of Protection Works and river Training Works complete in all respects~~	-	~~**Construction of Balance Protection Work and River Training Works:** On completion of Protection and river Training Works (parapet, slope protection, wearing coat) complete in all respects as specified.~~

1.3.3 Other works.

Procedure for estimating the value of other works done shall be as stated in table 1.3.4.

Table 1.3.4

Stage of Payment	Percentage Weightage	Payment Procedure
Miscellaneous:		
(i) RCC Encased Pipe Drain with Paved Footpath Over It for built up area and junction	**59.93%**	Unit of measurement is linear length in km. Payment shall be made on pro rata basis on completion of a stage of 200.00 meter length in a complete.
(ii) Barrier Type Kerb	**1.81%**	Unit of measurement is linear length in km. Payment shall be made on pro rata basis on completion of a stage of 200.00 meter length in a complete.
(iii) Road Markings, Road Signs, Studs, Solar Blinker, Rumble Strips Crash Barrier, Delineators, Safety Boards, Hazard Markers, etc.	**1.66%**	Unit of measurement is linear length in km. Payment shall be made on pro rata basis on completion of a stage in a complete.
(iv) Street Lighting& High Mast	**8.54%**	Cost of each street light shall be determined on pro rata basis with respect to the total number of street lights & high mast. Payment shall be made on the completion of all facilities.
(v)Junctions Development	**14.24%**	Payment shall be made on pro rata basis for completed facilities.
(vi) Paver Block including Crust under it, complete	**13.08%**	Payment shall be made on pro rata basis for completed facilities.
(vii) Utility Pipe Ducts	**0.74%**	Payment shall be made on pro rata basis for completed facilities.

1.3.4 Electrical Utilities

Table 1.3.4

Stage of payment	Weightage	Payment procedure
Electrical Utility Shifting	**12.41 %**	Payment shall be made on pro rata basis for completed facilities.
LT U/G Line		
11 KV U/G Line		
DP Structure with AB Switch		
SPCP (for Cable support)		
Shifting of 200 KVA Transformers		
Shifting of 100 KVA Transformers		

1.3.5 Water Utilities

Stage of payment	Weightage	Payment procedure
Water Utility Shifting	**6.52 %**	Payment shall be made on pro rata basis for completed facilities.
Shifting of CI Pipe		
Shifting of HDPE Pipe		

***The Contractor shall produce quantity & quality certification from the Competent Authority of Utility owning entity with the payment certificates submitted by the Contractor. The final release of payment against the shifting of utility will be made to the Contractor only after successful handing over of shifted utilities to the respective utility owning entity and after production of handing over taking over document to the Authority.**

2. Procedure for payment for Maintenance

NIL

SCHEDULE - I

(See Clause 10.2.4)

I. DRAWINGS

1. Drawings

In compliance of the obligations set forth in Clause 10.2 of this Agreement, the Contractor shall furnish to the Authority's Engineer, free of cost, all Drawings listed in Annex-I of this Schedule-I.

2. Additional Drawings

If the Authority's Engineer determines that for discharging its duties and functions under this Agreement, it requires any drawings other than those listed in Annex-I, it may by notice require the Contractor to prepare and furnish such drawings forthwith. Upon receiving a requisition to this effect, the Contractor shall promptly prepare and furnish such drawings to the Authority's Engineer, as if such drawings formed part of Annex-I of this Schedule-I.

Annex – I: List of Drawings

(Schedule - I)

a) Working drawings of all the components/ elements of the project highway as determined by Authority Engineer/NHAI and

b) As-built drawings for the project highway components / elements as determined by the Authority Engineer/NHAI. As built drawings shall be duly certified by NHAI.

A broad list of the drawings of the various components/elements of the project Highway and project facilities required to be submitted by the contractor is given below:.

I. Drawings of horizontal alignment, vertical profile and cross sections.

II. Drawings of junctions/intersections.

III. Drawings of Bridges, Culverts and CD structure

IV. Drawings of drainage (plan, profile, Invert levels, lead offs, etc.,)

V. Drawings of road furniture and traffic appurtenances.

VI. Drawings of traffic diversions and road safety.

VII. Drawings of urban areas, footpaths, drains, pedestrian crossings, etc.

VIII. Drawings of lighting along the project road.

IX. Drawings of rain water harvesting systems.

X. Drawings of retaining walls and protection works.

XI. Any other drawings as deemed necessary by NHAI/AE.

SCHEDULE - J

(See Clause 10.3.2)

J. PROJECT COMPLETION SCHEDULE

1. Project Completion Schedule

During Construction period, the Contractor shall comply with the requirements set forth in this Schedule-J for each of the Project Milestones and the **Scheduled Completion Date**. Within 15 (fifteen) days of the date of each Project Milestone, the Contractor shall notify the Authority of such compliance along with necessary particulars thereof.

2. Project Milestone-I

2.1 Project Milestone-I shall occur on the date falling on **the [90 Days of the Scheduled Construction Period]** day from the Appointed Date **(the "Project Milestone- I").**

2.2 Prior to the occurrence of Project Milestone-I, the Contractor shall have commenced construction of the Project Highway and submitted to the Authority duly and validly prepared Stage Payment Statements for an amount not less than **10%** of the Contract Price.

3. Project Milestone-II

3.1 Project Milestone-II shall occur on the date falling on the **[150 Days of the Scheduled Construction Period]** day from the Appointed Date **(the "Project Milestone- II").**

3.2 Prior to the occurrence of Project Milestone-II, the Contractor shall have continued with construction of the Project Highway and submitted to the Authority duly and validly prepared Stage Payment Statements for an amount not less than **35%** of the Contract Price and should have started construction of all bridges.

4. Project Milestone-III

4.1 Project Milestone-III shall occur on the date falling on the **[210 Days of the Scheduled Construction Period]** day from the Appointed Date **(the "Project Milestone- III").**

4.2 Prior to the occurrence of **Project Milestone-III**, the Contractor shall have continued with construction of the Project Highway and submitted to the Authority duly and validly prepared Stage Payment Statements for an amount not less than **70%** of the Contract Price and should have started construction of all project facilities.

5. Scheduled Completion Date

5.1 The Scheduled Completion Date shall occur on the **[270 Days]** from the Appointed Date.

5.2 On or before the Scheduled Completion Date, the Contractor shall have completed construction in accordance with this Agreement.

6. Extension of time

Upon extension of any or all of the aforesaid Project Milestones or the Scheduled Completion Date, as the case may be, under and in accordance with the provisions of this Agreement, the Project Completion Schedule shall be deemed to have been amended accordingly.

Schedule K

(See Clause 12.1.2)

K. TESTS ON COMPLETION

1. Schedule for Tests

1.1 The Contractor shall, no later than 30 (thirty) days prior to the likely completion of construction, notify the Authority's Engineer and the Authority of its intent to subject the Project Highway to Tests, and no later than 10 (ten) days prior to the actual date of Tests, furnish to the Authority's Engineer and the Authority detailed inventory and particulars of all works and equipment forming part of Works.

1.2 The Contractor shall notify the Authority's Engineer of its readiness to subject the Project Highway to Tests at any time after 10 (ten) days from the date of such notice, and upon receipt of such notice, the Authority's Engineer shall, in consultation with the Contractor, determine the date and time for each Test and notify the same to the Authority who may designate its representative to witness the Tests. The Authority's Engineer shall thereupon conduct the Tests itself or cause any of the Tests to be conducted in accordance with Article 12 and this Schedule-K.

2. Tests

2.1 Visual and physical test: The Authority's Engineer shall conduct a visual and physical check of construction to determine that all works and equipment forming part thereof conform to the provisions of this Agreement. The physical tests shall include all the tests required for quality control or as decided with authority's engineer at the time of physical tests as per relevant IRC and MORTH specifications

2.2 Riding quality test: Riding quality of each lane of the carriageway shall be checked with the help of a calibrated bump integrator and the maximum permissible roughness for purposes of this Test shall be [2,000 (two thousand)] mm for each kilometer.

2.3 Tests for bridges: All major and minor bridges shall be subjected to the rebound hammer and ultrasonic pulse velocity tests, to be conducted in accordance with the procedure described in Special Report No. 17: 1996 of the IRC Highway Research Board on Nondestructive Testing Techniques, at two spots in every span, to be chosen at random by the Authority's Engineer. Bridges with a span of 15 (fifteen) meters or more shall also be subjected to load testing.

2.4 Pavement Strength: The pavement strength is to be measured using Falling Weight Deflectometer (FWD), at least once a year upon Completion of Project.

2.5 Other tests: The Authority's Engineer may require the Contractor to carry out or cause to be carried additional tests, in accordance with Good Industry Practice, for determining the compliance of the Project Highway with Specifications and Standards, except tests as specified in clause 5,but shall include measuring the reflectivity of road markings and road signs; and measuring the illumination level (lux) of lighting using requisite testing equipment.

2.6 Environmental audit: The Authority's Engineer shall carry out a check to determine conformity of the Project Highway with the environmental requirements set forth in Applicable Laws and Applicable Permits.

2.7 Safety Audit: The Authority's Engineer shall carry out, or cause to be carried out, a safety audit to determine conformity of the Project Highway with the safety requirements and Good Industry Practice.

3. Agency for conducting Tests

~~All Tests set forth in this Schedule-K shall be conducted by the Authority's Engineer or such other agency or person as it may specify in consultation with the Authority.~~

4. Completion Certificate

Upon successful completion of Tests, the Authority's Engineer shall issue the Completion Certificate in accordance with the provisions of Article 12.

5. Tests By Authority Engineer

The Authority Engineer will witness tests with following equipment in the presence of contractor's representative

~~Sr. No.~~	~~Key metrics of Asset~~	~~Equipment to be used~~	~~Frequency of condition survey~~
~~1~~	~~Surface defectsof pavement~~	~~Network Survey Vehicle(NSV)~~	~~At least twice a year (As per survey months defined for the state basis rainy season)~~
~~2~~	~~Roughness of pavement~~	~~Network Survey Vehicle(NSV)~~	~~At least twice a year (As per survey months defined for the state basis rainy season)~~
~~3~~	~~Strength of pavement~~	~~Falling Weight Deflectometer (FWD)~~	~~At least once a year~~
~~4~~	~~Bridges~~	~~Mobile Bridge Inspection Unit (MBU)~~	~~At least twice a year (As per survey months defined for the state basis rainy season)~~
5	Road signs	Retro-Reflectometer	At least Twice a year

Note :- The first testing with the help of NSV shall be conducted at the time of issue of Completion Certificate.

SCHEDULE - L

(See Clause 12.2 and 12.4)

L. CERTIFICATES

1. Completion Certificate

1. I, ……. (Name of the Authority's Engineer), acting as the Authority's Engineer, under and in accordance with the Agreement dated…… (the "Agreement"), for "Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra on EPC Mode." through……. (Name of Contractor), hereby certify that the Tests in accordance with Article 12 of the Agreement have been successfully undertaken to determine compliance of the Project Highway with the provisions of the Agreement, and I am satisfied that the Project Highway can be safely and reliably placed in service of the Users thereof.

2. It is certified that, in terms of the aforesaid agreement, all works forming part of Project Highway have been completed, and the Project Highway is hereby declared fit for entry into operation on this the……day of…… 20….

SIGNED, SEALED AND DELIVERED

For and on behalf of the Authority's Engineer by:

(Signature)

(Name)

(Designation)

(Address)

SCHEDULE - M

(See Clauses 14.6, 15.2 and 19.7)

M. PAYMENT REDUCTION FOR NON-COMPLIANCE

1. Payment reduction for non-compliance with the Maintenance Requirements

1.1 Monthly lump sum payments for maintenance shall be reduced in the case of non-compliance with the Maintenance Requirements set forth in Schedule-E.

1.2 Any deduction made on account of non-compliance with the Maintenance Requirements shall not be paid even after compliance subsequently. The deductions shall continue to be made every month until compliance is done.

1.3 The Authority's Engineer shall calculate the amount of payment reduction on the basis of weightage in percentage assigned to non-conforming items as given in Paragraph 2.

2. Percentage reductions in lump sum payments

2.1 The following percentages shall govern the payment reduction:

Sl. No.	Item/Defect/Deficiency	Percentage
(a)	**Carriageway/Pavement**	
(i)	Potholes, cracks, other surface Defects	15%
(ii)	Repairs of Edges, Rutting	5%
(b)	**Road, Embankment, Cuttings,** **Shoulders**	
(i)	Edge drop, inadequate crossfall, undulations, settlement, potholes, ponding, obstructions	10%
(ii)	Deficient slopes, raincuts, disturbed pitching, vegetation growth, pruning of trees	5%
(c)	**Bridges and Culverts**	
(i)	Desilting, cleaning. vegetation growth, damaged pitching, flooring, parapets, wearing course, footpaths, any damage to foundations	20%
(ii)	Any Defects in superstructures, bearings and sub-structures	10%
(iii)	Painting, repairs/replacement kerbs, railings, parapets, guideposts/crash barriers	5%
(d)	**Roadside Drains**	

Sl. No.	Item/Defect/Deficiency	Percentage
(i)	Cleaning and repair of drains	5%
(e)	**Road Furniture**	
(i)	Cleaning, painting, replacement of road signs, delineators, road markings, 200 m/km/5th km stones	5%
(f)	**Miscellaneous Items**	
(i)	Removal of dead animals, broken down/accidented vehicles, fallen trees, road blockades or malfunctioning of mobile crane	10%
(ii)	Any other Defects in accordance with paragraph 1.	5%
(g)	**Defects in Other Project Facilities**	5%

2.2 The amount to be deducted from monthly lump-sum payment for noncompliance of particular item shall be calculated as under:

$$R=P/100 \times M \times L1/L$$

Where,

P = Percentage of particular item/Defect/deficiency for deduction

M = Monthly lump-sum payment in accordance with the Bid

L1 = Non-complying length L = Total length of the road,

R = Reduction (the amount to be deducted for noncompliance for a particular item/Defect/deficiency

The total amount of reduction shall be arrived at by summation of reductions for such items/Defects/deficiency or noncompliance.

For any Defect in a part of one kilometer, the non-conforming length shall be taken as one kilometer.

SCHEDULE - N

(See Clause 18.1.1)

N. SELECTION OF AUTHORITY'S ENGINEER

(State PWD Field Unit will be Authority's Engineer for said project)

1. Selection of Authority's Engineer

1.1 The provisions of the Model Request for Proposal for Selection of Technical Consultants, issued by the Ministry of Finance in May 2009, or any substitute thereof shall apply for selection of an experienced firm to discharge the functions and duties of an Authority's Engineer.

1.2 In the event of termination of the Technical Consultants appointed in accordance with the provisions of Paragraph 1.1, the Authority shall appoint another firm of Technical Consultants forthwith and may engage a government-owned entity in accordance with the provisions of Paragraph 3 of this Schedule-N.

2. Terms of Reference

The Terms of Reference for the Authority's Engineer (the "TOR") shall substantially conform with Annex 1 to this Schedule N.

3. Appointment of Government entity as Authority's Engineer

Notwithstanding anything to the contrary contained in this Schedule, the Authority may in its discretion appoint a government-owned entity as the Authority's Engineer; provided that such entity shall be a body corporate having as one of its primary functions the provision of consulting, advisory and supervisory services for engineering projects; provided further that a government-owned entity which is owned or controlled by the Authority shall not be eligible for appointment as Authority's Engineer.

Annex – I: TERMS OF REFERENCE FOR AUTHORITY'S ENGINEER

(Schedule - N)

1. Scope

1.1 These Terms of Reference (the "**TOR**") for the Authority's Engineer are being specified pursuant to the EPC Agreement dated..... (The "**Agreement**), which has been entered into between the [name and address of the Authority] (the "**Authority**") and..... (the "**Contractor**") for "Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra on EPC Mode." and a copy of which is annexed hereto and marked as Annex-A to form part of this TOR.

1.2 The TOR shall apply to construction and maintenance of the Project Highway.

2. Definitions and interpretation

2.1 The words and expressions beginning with or in capital letters and not defined herein but defined in the Agreement shall have, unless repugnant to the context, the meaning respectively assigned to them in the Agreement.

2.2 References to Articles, Clauses and Schedules in this TOR shall, except where the context otherwise requires, be deemed to be references to the Articles, Clauses and Schedules of the Agreement, and references to Paragraphs shall be deemed to be references to Paragraphs of this TOR.

2.3 The rules of interpretation stated in Clauses 1.2, 1.3 and 1.4 of the Agreement shall apply, mutatis mutandis, to this TOR.

3. General

3.1 The Authority's Engineer shall discharge its duties in a fair, impartial and efficient manner, consistent with the highest standards of professional integrity and Good Industry Practice.

3.2 The Authority's Engineer shall perform the duties and exercise the authority in accordance with the provisions of this Agreement, but subject to obtaining prior written approval of the Authority before determining:

(a) Any Time Extension;

(b) Any additional cost to be paid by the Authority to the Contractor;

(c) The Termination Payment; or

(d) Issuance of Completion Certificate or

(e) any other matter which is not specified in (a), (b), (c) or (d) above and which creates a financial liability on either Party

3.3 The Authority's Engineer shall submit regular periodic reports, at least once every month, to the Authority in respect of its duties and functions under this Agreement. Such reports shall be submitted by the Authority's Engineer within 10 (ten) days of the beginning of every month.

3.4 The Authority's Engineer shall ensure that NSV survey shall be conducted at the following intervals: (13.11.2019)

Before start of the work,

Before issue of provisional/ final completion certificate,

Every 6 months after completion of work

The NSV reports shall be uploaded on PMIS and RAMS portal (http://nsvsurvey.nhai.org/NSVsurvey/)

3.5 The Authority's Engineer in consultation with the Authority shall update PMIS Portal of the Ministry (https://pmis.morthedisha.gov.in/) and also ensure uploading Monthly and Quarterly Progress Report along with Strip Chart on PMIS Portal. Such updations shall be done on the last date of every month/ quarter and if required, on a weekly basis as advised by the Authority. Monthly invoices towards services of the Authority's Engineer shall not be admitted for payment in case of non-compliance of the above.

3.6 The Authority's Engineer shall inform the Contractor of any delegation of its duties and responsibilities to its suitably qualified and experienced personnel; provided, however, that it shall not delegate the authority to refer any matter for the Authority's prior approval in accordance with the provisions of Clause 18.2.

3.7 The Authority's Engineer shall aid and advise the Authority on any proposal for Change of Scope under Article 13.

3.8 In the event of any disagreement between the Parties regarding the meaning, scope and nature of Good Industry Practice, as set forth in any provision of the Agreement, the Authority's Engineer shall specify such meaning, scope and nature by issuing a reasoned written statement relying on good industry practice and authentic literature.

4. Construction Period

4.1 During the Construction Period, the Authority's Engineer shall review the Drawings furnished by the Contractor along with supporting data, including the geo-technical and hydrological investigations, characteristics of materials from borrow areas and quarry sites, topographical surveys, and the recommendations of the Safety Consultant in accordance with the provisions of Clause 10.1.6. The Authority's Engineer shall complete such review and send its observations to the Authority and the Contractor within 15

(fifteen) days of receipt of such Drawings; provided, however that in case of a Major Bridge or Structure, the aforesaid period of 15 (fifteen) days may be extended upto 30 (thirty) days. In particular, such comments shall specify the conformity or otherwise of such Drawings with the Scope of the Project and Specifications and Standards.

4.2 The Authority's Engineer shall review any revised Drawings sent to it by the Contractor and furnish its comments within 10 (ten) days of receiving such Drawings.

4.3 The Authority's Engineer shall review the Quality Assurance Plan submitted by the Contractor and shall convey its comments to the Contractor within a period of 21 (twenty-one) days stating the modifications, if any, required thereto.

4.4 The Authority's Engineer shall complete the review of the methodology proposed to be adopted by the Contractor for executing the Works, and convey its comments to the Contractor within a period of 10 (ten) days from the date of receipt of the proposed methodology from the Contractor.

4.5 The Authority's Engineer shall grant written approval to the Contractor, where necessary, for interruption and diversion of the flow of traffic in the existing lane(s) of the Project Highway for purposes of maintenance during the Construction Period in accordance with the provisions of Clause 10.4.

4.6 The Authority's Engineer shall review the monthly progress report furnished by the Contractor and send its comments thereon to the Authority and the Contractor within 7 (seven) days of receipt of such report.

4.7 The Authority's Engineer shall inspect the Construction Works and the Project Highway and shall submit a monthly Inspection Report bringing out the results of inspections and the remedial action taken by the Contractor in respect of Defects or deficiencies. In particular, the Authority's Engineer shall include in its Inspection Report, the compliance of the recommendations made by the Safety Consultant.

4.8 The Authority's Engineer shall conduct the pre-construction review of manufacturer's test reports and standard samples of manufactured Materials, and such other Materials as the Authority's Engineer may require.

4.9 For determining that the Works conform to Specifications and Standards, the Authority's Engineer shall require the Contractor to carry out, or cause to be carried out, tests at such time and frequency and in such manner as specified in the Agreement and in accordance with Good Industry Practice for quality assurance. For purposes of this Paragraph 4.9, the tests specified in the IRC Special Publication-11 (Handbook of Quality Control for Construction of Roads and Runways) and the Specifications for Road and Bridge Works issued by MOSRTH (the "Quality Control Manuals") or any modification/substitution

thereof shall be deemed to be tests conforming to Good Industry Practice for quality assurance.

4.10 The Authority's Engineer shall test check at least 20 (twenty) percent of the quantity or number of tests prescribed for each category or type of test for quality control by the Contractor.

4.11 The timing of tests referred to in Paragraph 4.9, and the criteria for acceptance/ rejection of their results shall be determined by the Authority's Engineer in accordance with the Quality Control Manuals. The tests shall be undertaken on a random sample basis and shall be in addition to, and independent of, the tests that may be carried out by the Contractor for its own quality assurance in accordance with Good Industry Practice.

4.12 In the event that results of any tests conducted under Clause 11.10 establish any Defects or deficiencies in the Works, the Authority's Engineer shall require the Contractor to carry out remedial measures.

4.13 The Authority's Engineer may instruct the Contractor to execute any work which is urgently required for the safety of the Project Highway, whether because of an accident, unforeseeable event or otherwise; provided that in case of any work required on account of a Force Majeure Event, the provisions of Clause 21.6 shall apply.

4.14 In the event that the Contractor fails to achieve any of the Project Milestones, the Authority's Engineer shall undertake a review of the progress of construction and identify potential delays, if any. If the Authority's Engineer shall determine that completion of the Project Highway is not feasible within the time specified in the Agreement, it shall require the Contractor to indicate within 15 (fifteen) days the steps proposed to be taken to expedite progress, and the period within which the Project Completion Date shall be achieved. Upon receipt of a report from the Contractor, the Authority's Engineer shall review the same and send its comments to the Authority and the Contractor forthwith.

4.15 The Authority's Engineer shall obtain from the Contractor a copy of all the Contractor's quality control records and documents before the Completion Certificate is issued pursuant to Clause 12.4.

4.16 Authority's Engineer may recommend to the Authority suspension of the whole or part of the Works if the work threatens the safety of the Users and pedestrians. After the Contractor has carried out remedial measure, the Authority's Engineer shall inspect such remedial measures forthwith and make a report to the Authority recommending whether or not the suspension hereunder may be revoked.

4.17 In the event that the Contractor carries out any remedial measures to secure the safety of suspended works and Users, and requires the Authority's Engineer to inspect such works,

the Authority's Engineer shall inspect the suspended works within 3 (three) days of receiving such notice, and make a report to the Authority forthwith, recommending whether or not such suspension may be revoked by the Authority.

4.18 The Authority's Engineer shall carry out, or cause to be carried out, all the Tests specified in Schedule-K and issue a Completion Certificate or Provisional Certificate, as the case may be. For carrying out its functions under this Paragraph 4.18 and all matters incidental thereto, the Authority's Engineer shall act under and in accordance with the provisions of Article 12 and Schedule-K.

5. Maintenance Period

5.1 The Authority's Engineer shall aid and advise the Contractor in the preparation of its monthly Maintenance Programme and for this purpose carry out a joint monthly inspection with the Contractor.

5.2 The Authority's Engineer shall undertake regular inspections, at least once every month, to evaluate compliance with the Maintenance Requirements and submit a Maintenance Inspection Report to the Authority and the Contractor.

5.3 The Authority's Engineer shall specify the tests, if any, that the Contractor shall carry out, or cause to be carried out, for the purpose of determining that the Project Highway is in conformity with the Maintenance Requirements. It shall monitor and review the results of such tests and the remedial measures, if any, taken by the Contractor in this behalf.

5.4 In respect of any defect or deficiency referred to in Paragraph 3 of Schedule- E, the Authority's Engineer shall, in conformity with Good Industry Practice, specify the permissible limit of deviation or deterioration with reference to the Specifications and Standards and shall also specify the time limit for repair or rectification of any deviation or deterioration beyond the permissible limit.

5.5 The Authority's Engineer shall examine the request of the Contractor for closure of any lane(s) of the Project Highway for undertaking maintenance/repair thereof, and shall grant permission with such modifications, as it may deem necessary, within 5 (five) days of receiving a request from the Contractor. Upon expiry of the permitted period of closure, the Authority's Engineer shall monitor the reopening of such lane(s), and in case of delay, determine the Damages payable by the Contractor to the Authority under Clause 14.5.

6. Determination of costs and time

6.1 The Authority's Engineer shall determine the costs, and/or their reasonableness, that are required to be determined by it under the Agreement.

6.2 The Authority's Engineer shall determine the period of Time Extension that is required to be determined by it under the Agreement.

6.3 The Authority's Engineer shall consult each Party in every case of determination in accordance with the provisions of Clause 18.5.

7. Payments

7.1 The Authority's Engineer shall withhold payments for the affected works for which the Contractor fails to revise and resubmit the Drawings to the Authority's Engineer in accordance with the provisions of Clause 10.2.4 (d).

7.2 Authority's Engineer shall -

(a) within 10 (ten) days of receipt of the Stage Payment Statement from the Contractor pursuant to Clause 19.4, determine the amount due to the Contractor and recommend the release of 90 (ninety) percent of the amount so determined as part payment, pending issue of the Interim Payment Certificate; and

(b) within 15 (fifteen) days of the receipt of the Stage Payment Statement referred to in Clause 19.4, deliver to the Authority and the Contractor an Interim Payment Certificate certifying the amount due and payable to the Contractor, after adjustments in accordance with the provisions of Clause 19.10.

7.3 The Authority's Engineer shall, within 15 (fifteen) days of receipt of the Monthly Maintenance Statement from the Contractor pursuant to Clause 19.6, verify the Contractor's monthly statement and certify the amount to be paid to the Contractor in accordance with the provisions of the Agreement.

7.4 The Authority's Engineer shall certify final payment within 30 (thirty) days of the receipt of the final payment statement of Maintenance in accordance with the provisions of Clause 19.16.

8. Other duties and functions

The Authority's Engineer shall perform all other duties and functions as specified in the Agreement.

9. Miscellaneous

9.1 A copy of all communications, comments, instructions, Drawings or Documents sent by the Authority's Engineer to the Contractor pursuant to this TOR, and a copy of all the test results with comments of the Authority's Engineer thereon, shall be furnished by the Authority's Engineer to the Authority forthwith.

9.2 The Authority's Engineer shall retain at least one copy each of all Drawings and Documents received by it, including 'as-built' Drawings, and keep them in its safe custody.

9.3 Within 90 (ninety) days of the Project Completion Date, the Authority's Engineer shall obtain a complete set of as-built Drawings, in 2 (two) hard copies and in micro film form or in such other medium as may be acceptable to the Authority, reflecting the Project

Highway as actually designed, engineered and constructed, including an as-built survey illustrating the layout of the Project Highway and setback lines, if any, of the buildings and structures forming part of Project Facilities; and shall hand them over to the Authority against receipt thereof.

9.4 The Authority's Engineer, if called upon by the Authority or the Contractor or both, shall mediate and assist the Parties in arriving at an amicable settlement of any Dispute between the Parties.

9.5 The Authority's Engineer shall inform the Authority and the Contractor of any event of Contractor's Default within one week of its occurrence.

SCHEDULE - O

(See Clauses 19.4.1, 19.6.1, and 19.8.1)

O. Forms of Payment Statements

1. Stage Payment Statement for Works

The Stage Payment Statement for Works shall state:

(a) The estimated amount for the Works executed in accordance with Clause 19.3.1 subsequent to the last claim;

(b) Amounts reflecting adjustments in price for the aforesaid claim;

(c) The estimated amount of each Change of Scope Order executed subsequent to the last claim;

(d) Amounts reflecting adjustment in price, if any, for (c) above in accordance with the provisions of Clause 13.2.3 (a);

(e) Total of (a), (b), (c) and (d) above;

(f) Deductions:

(i) Any amount to be deducted in accordance with the provisions of the Agreement except taxes;

(ii) Any amount towards deduction of taxes; and

(iii) Total of (i) and (ii) above.

(g) Net claim: (e) – (f) (iii);

(h) The amounts received by the Contractor up to the last claim:

(i) For the Works executed (excluding Change of Scope orders);

(ii) For Change of Scope Orders, and

(iii) Taxes deducted

2. Monthly Maintenance Payment Statement

The monthly Statement for Maintenance Payment shall state:

(i) The monthly payment admissible in accordance with the provisions of the Agreement;

(ii) The deductions for maintenance work not done;

(iii) Net payment for maintenance due, (a) minus (b);

(iv) Amounts reflecting adjustments in price under clause 19.12; and

(v) Amount towards deduction of taxes

3. Contractor's claim for Damages

Note: The Contractor shall submit its claims in a form acceptable to the Authority.

SCHEDULE - P

(See Clause 20.1)

P. INSURANCE

1. Insurance during Construction Period

1.1 The Contractor shall effect and maintain at its own cost, from the Appointed Date till the date of issue of the Completion Certificate, the following insurances for any loss or damage occurring on account of Non Political Event of Force Majeure, malicious act, accidental damage, explosion, fire and terrorism:

(a) Insurance of Works, Plant and Materials and an additional sum of [15 (fifteen)] per cent of such replacement cost to cover any additional costs of and incidental to the rectification of loss or damage including professional fees and the cost of demolishing and removing any part of the Works and of removing debris of whatsoever nature; and

(b) Insurance for the Contractor's equipment and Documents brought onto the Site by the Contractor, for a sum sufficient to provide for their replacement at the Site.

1.2 The insurance under paragraph 1.1 (a) and (b) above shall cover the Authority and the Contractor against all loss or damage from any cause arising under paragraph 1.1 other than risks which are not insurable at commercial terms.

2. Insurance for Contractor's Defects Liability

The Contractor shall effect and maintain insurance cover for the Works from the date of issue of the Completion Certificate until the end of the Defects Liability Period for any loss or damage for which the Contractor is liable and which arises from a cause occurring prior to the issue of the Completion Certificate. The Contractor shall also maintain other insurances for maximum sums as may be required under the Applicable Laws and in accordance with Good Industry Practice.

3. Insurance against injury to persons and damage to property

3.1 The Contractor shall insure against its liability for any loss, damage, death or bodily injury, or damage to any property (except things insured under Paragraphs 1 and 2 of this Schedule or to any person (except persons insured under Clause 20.9), which may arise out of the Contractor's performance of this Agreement. This insurance shall be for a limit per occurrence of not less than the amount stated below with no limit on the number of occurrences.

The insurance cover shall be not less than: Rs. [*****]

The insurance shall be extended to cover liability for all loss and damage to the Authority's property arising out of the Contractor's performance of this Agreement excluding:

(a) The Authority's right to have the construction works executed on, over, under, in or through any land, and to occupy this land for the Works; and

(b) Damage which is an unavoidable result of the Contractor's obligations to execute the Works.

4. Insurance to be in joint names

The insurance under paragraphs 1 to 3 above shall be in the joint names of the Contractor and the Authority.

SCHEDULE - Q

(See Clause 20.1)

Q. TESTS ON COMPLETION OF MAINTENANCE PERIOD

1. Riding Quality test:

Riding quality of each lane of the carriageway shall be checked with the help of a calibrated bump integrator and the maximum permissible roughness for purposes of this Test shall be [2,200 (two thousand and two hundred only)] mm for each kilometer.

2. Visual and physical test:

The Authority's Engineer shall conduct a visual and physical check of construction to determine that all works and equipment forming part thereof conform to the provisions of this Agreement. The physical tests shall include measurement of cracking, rutting, stripping and potholes and shall be as per the requirement of maintenance mentioned in Schedule-E.

SCHEDULE - R

(See Clause 14.10)

R. TAKING OVER CERTIFICATE

I, ……. (Name and designation of the Authority's Representative) under and in accordance with the Agreement dated…… (the "**Agreement**"), for "Upgradation of BSNL Office to Railway Crossing in Bhandara City from Chainage Km 168+180 to Km 168+920 (Length - 0.740 Km) of NH-247 to Two Lane with Paved Shoulder Configuration of Bhandara-Khat-Ghotitok road in the state of Maharashtra on EPC Mode." through ……. (Name of Contractor), hereby certify that the Tests on completion of Maintenance Period in accordance with Article 14 of the Agreement have been successfully undertaken to determine compliance of the Project Highway with the provisions of the Agreement and I hereby certify that the Authority has taken over the Project highway from the Contractor on this day.....

SIGNED, SEALED ANDDELIVERED

(Signature)

(Name and designation of Authority'sRepresentative)

(Address)

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